Automatic Composite-Modulation Classification Using Cyclic-Paw-Print Features for Cognitive Aerospace Communications

Xiao Yan, *Member, IEEE*, Xunuo Zhong, Hsiao-Chun Wu, *Fellow, IEEE*, PengFei Yang, Qian Wang, and Yiyun Chen

***Abstract*— Automatic composite-modulation classification (ACMC) is deemed to be an essential and important cognitive mechanism adopted for the next generation intelligent telemetry, tracking, and command (TT&C), cognitive aerospace communications, and space surveillance as it can automatically recognize the unknown composite-modulation (CM) scheme of the received signal. In this work, we introduce a novel ACMC method using the hybrid feature-vectors based on our proposed new cyclic-paw-print images extracted from the CM signals. In our proposed novel approach, according to the polyspectral analysis of the received CM signals, a received CM signal can be converted to a gray-scale image matrix, named cyclic-paw-print (CPP), which can be very robust against noise. Then, the discrete cosine transform (DCT) and discrete wavelet transform (DWT) are simultaneously applied to the obtained CPP matrix and the hybrid DCT/DWT feature-vector is further constructed thereby. Finally, the sequential minimal optimized support vector machine (SMO-SVM) is adopted as the classifier using such feature vectors. Our proposed new ACMC technique requires a lower computational complexity and leads to a higher recognition-accuracy than other existing ACMC methods according to Monte Carlo simulations and experiments using real CM signal data.**

***Index Terms*— Intelligent telemetry, tracking, and command (TT&C) systems, automatic composite-modulation classification (ACMC), cyclic-paw-print (CPP), discrete cosine transform (DCT), discrete wavelet transform (DWT), sequential minimal optimized support vector machine (SMO-SVM).**

Manuscript received 1 June 2023; revised 8 November 2023 and 23 February 2024; accepted 6 April 2024. Date of publication 15 April 2024; date of current version 18 September 2024. This work was supported in part by the Natural Science Foundation of Sichuan Province under Grant 2022NSFSC0545 and 2023NSFSC0494, and the Louisiana Board of Regents Research Competitiveness Subprogram under Grant LEQSF(2021-22)-RD-A-34. The associate editor coordinating the review of this article and approving it for publication was M. A. Imran. *(Corresponding author: Xiao Yan.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCOMM.2024.3388509.

Digital Object Identifier 10.1109/TCOMM.2024.3388509

I. Introduction

COMPOSITE modulation (CM), which is referred to as the advanced multi-layer modulation scheme for multi-service base-band data using a plurality of modulation schemes, has been considered as a critical technology to enhance the transmission efficiency, anti-interference and anti-interception capability, and security in cognitive aerospace communications [1], [2], [3], [4], [5]. CM has been widely employed in aerospace vehicles to facilitate 6G (sixth generation) communication networks, spacecraft and terrestrial communications, and space sensing [6], [7], [8]. The recent emergence of low-earth orbit (LEO) satellite broadband internet and drones leads to the imperious demand of cognitive radio technology for boosting the spectrum efficiency and mitigating the interference by spectrum sensing and sharing, and the multi-layer modulation schemes of the received CM signals have to be blindly identified beforehand [9], [10], [11]. Therefore, *automatic composite-modulation classification* (ACMC), namely to autonomously recognize the composite-modulation (CM) scheme of an unknown received signal without any *a priori* knowledge, can be deemed an important and promising mechanism in facilitating the 6G communication systems, cognitive space radios, smart Internet of Things (IoT), and space surveillance [12], [13], [14], [15].

The related problem of the ACMC problem is the well-known automatic modulation classification (AMC) problem. Almost all conventional automatic modulation classification (AMC) methods cannot be applied for ACMC as they are under the universal assumption that the received signal results from a single-layer modulation scheme. Due to the nonlinear coupling between the multi-layer modulations in a CM signal, the commonly-adopted likelihood functions and features in the existing AMC methods (see [16], [17], [18], [19], [20], [21], [22], [23]) cannot lead to satisfactory recognition-accuracies for ACMC [24]. Consequently, new robust and efficient ACMC techniques are surely of great interest.

As a matter of fact, the ACMC problem has rarely been tackled so far. The common but intuitive ACMC approach is the "*layer-by-layer identification and demodulation* (LI&D)" approach, i.e., the receiver needs to recognize the outer-layer modulation type of a received CM signal using one of the existing AMC methods and then demodulate it for the further inner-layer modulation identification [22], [25], [26], [27],

[28], [29]. However, the existing LI&D approach has the serious restriction that it would suffer from variants such as phase and timing offsets and high implementation complexity [30]. The amplitude spectrum and power spectrum of the received signal were utilized as the features and the support vector machines (SVM) was adopted as the classifier for ACMC in [31]. This method is unserviceable to CM signals classification in actual scenario for the introduction of pre-demodulating module. The two-fold phase-locked loop (PLL) was used to extract the CM signal features and the goodness-of-fit test was employed to measure the similarities between the actual and the pre-modeled phase cumulative distributions for ACMC [25]. This method would not be effective if the PLL fails to lock the correct phase. On the other hand, the CM scheme of the received signal can also be directly identified by the features utilized in the conventional AMC approaches, including instantaneous-amplitude characteristics (IAC), Fourier and power spectra, high-order statistics (HOSs), wavelet coefficients, cyclic spectra, etc. The ACMC method proposed in [32] extracted four features related to the HOSs and the square spectrum of the received CM signal and performed the ACMC using a decision tree. According to [32], the average recognition-accuracy for six CM candidate schemes can reach 98% in high signal-to-noise ratios (SNRs). However, its performance dramatically degraded when SNR slightly decreased and the sample size was required to be very large so as to attain reliable HOS estimates. The ACMC for radar waveforms with low probabilities of intercept (LPI) in the presence of noise was carried out using the features extracted from the fractional Fourier transform [33]. This method could achieve a promising classification performance by using the Wigner-Ville distribution, but its performance would dramatically degraded when it is applied to the CM signals employed in the telemetry, tracking, and command (TT&C) and space communications due to the mismatch of signal models. On the other hand, deep-learning (DL) networks have been employed for AMC, which can automatically learn the primary characteristics of communication signals [26], [27], [34], [35], [36], [37]. Various DL networks, including convolution neural networks (CNNs) in [38], [39], and [40], recurrent neural networks (RNNs) in [14] and [37], generative adversarial networks (GANs) in [41] and [42], and their variants, have been adopted for AMC. However, as the features are quite difficult to extract from CM signals, there hardly exists any DL network dedicated to ACMC to the best of our knowledge.

According to our recent study of the CM signals, we find that the top view of the normalized *second-order cyclic spectrum* (NSCS) of a CM signal demonstrates distinctive spectral features, which are very robust against noise and interference. We call the aforementioned top view of the NSCS of a CM signal "*Cyclic-Paw-Print*" (CPP). Motivated by this discovery, we design a new ACMC scheme using the hybrid feature vector integrating the discrete cosine transform (DCT) and discrete wavelet transform (DWT) of the CPP of the CM signal. According to our proposed new approach, the CPP is first obtained from the NSCS of the received CM signal. Then, the discrete cosine transform (DCT) and discrete wavelet transform (DWT) based on the *Haar wavelet* are simultaneously applied to the constructed CPP. Furthermore, the hybrid feature vector is produced from the low-frequency components of the two-dimensional DCT matrix and the principal components of the two-dimensional DWT matrix. Finally, according to the hybrid feature vectors extracted from the training CM signal data subject to various noise levels, the sequential minimal optimized (SMO) multi-class support vector machine (SVM) is adopted to identify the CM scheme of the received signal. Monte Carlo simulation results demonstrate that our proposed novel ACMC method outperform other existing techniques. Besides, our proposed ACMC method requires a quite low computational complexity at the test stage. The main contributions of this work can be summarized as follows.

- The novel digested representation (feature map) of the CM signals, namely CPP, is proposed to form robust CM signal features for ACMC.
- The hybrid feature vectors resulting from the DCT and DWT of the CPP matrix are proposed and they can be calculated with a low computational complexity.
- A SMO-SVM is adopted as the classifier to recognize the CM scheme of a received signal based on the hybrid feature vectors resulting from the training and test CM signal data.
- The performance of our proposed novel ACMC approach is demonstrated to be promising using real CM signal data produced by a vector signal generator.

The rest of this paper is organized as follows. Section II presents the preliminaries of CM signals and the corresponding signal model. Our proposed novel ACMC method is introduced in Section III. The performance evaluation of our proposed new ACMC technique is undertaken through Monte Carlo simulations and real experiments in Section IV. Finally, conclusion will be drawn in Section V.

Nomenclature: Scalars are denoted by Italianized alphabets such as a; vectors are denoted by alphabets with overhead arrow notions such as $\vec{A}$; matrices are denoted by alphabets with overhead tilde notions such as $\tilde{A}$; sets are denoted by blackboard bold alphabets such as $\mathbb{A}$. $\vec{A}^T$ and $\tilde{A}^T$ represent the transposes of a vector $\vec{A}$ and a matrix $\tilde{A}$, respectively. $\vec{A}^H$ and $\tilde{A}^H$ represent the conjugate-transposes (Hermitian adjoints) of a vector $\vec{A}$ and a matrix $\tilde{A}$, respectively. The set of all real numbers is denoted by $\mathbb{R}$, and the symbol $\jmath \overset{\text{def}}{=} \sqrt{-1}$ is reserved for representing the unit imaginary number throughout this paper. $\Re\{\ \}$ and $\Im\{\ \}$ returns the real and imaginary parts of the complex number enclosed by the braces, respectively. $\otimes$ denotes the Kronecker product.

II. Preliminaries and Signal Model of CM Signals

The TT&C technology, which provides radio-frequency (RF) links between a space vehicle and the facilities on the ground, performs three fundamental tasks: command (also known as telecommand, TC), telemetry (TM), and tracking (also referred to as ranging, RNG) to ensure that space vehicles operate properly [43]. The TC functions allow the mission operation center (MOC) to send commands through an earth-to-space data burst transmission channel to control

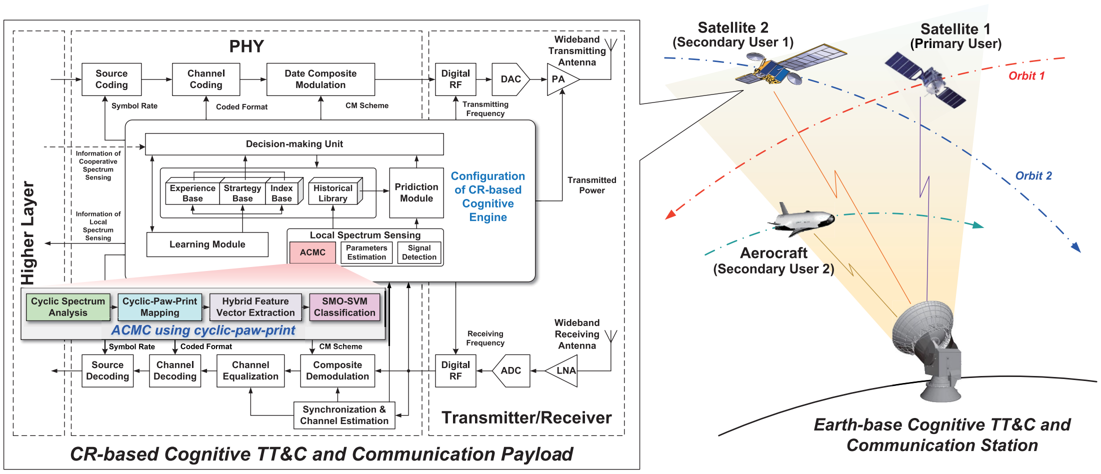

Fig. 1. The system diagram of the typical cognitive-radio-based cognitive TT&C and communication system.

a space vehicle. Meanwhile, the TM data, which consist of the spacecraft's recordings and status, are delivered to the ground through a space-to-earth link and then redirected to the MOC. CM is the essential modulation scheme involved in the aforementioned communications for TT&C. To improve the spectrum efficiency and mitigate the interference, cognitive radio technology has been already incorporated into the next cognitive TT&C and broadband space communication systems, where ACMC is employed by the secondary users to identify the CM scheme of the signal emitted by the primary user. According to [44], the typical cognitive-radio-based cognitive TT&C and communication system using ACMC is illustrated by Fig. 1, where the intelligent TT&C and communication payload carried by a space vehicle can sense its surrounding electromagnetic environment and follow the dynamic spectrum-allocation strategy to generate adaptive CM signals accordingly. Our proposed novel ACMC method in this work can be employed by the cognitive engine of the intelligent TT&C and communication payload to autonomously identify the CM scheme of the received signal. Moreover, ACMC has also been invoked to blindly recognize the modulation scheme of an unknown signal in space surveillance [44].

CM is a highly transmission-efficient modulation and multiplexing technique, where a primary carrier is modulated in conjunction with multiple subcarriers onto which multi-user data are multiplexed. It has been widely adopted by aerospace TT&C, space communications, and radar sensing. In this work, we focus on ACMC for two-layer CM schemes specified in the CCSDS standard (see [45]) for different space data-transmission applications. Two kinds of double-layer CM strategies are investigated here, where M-ary phase-shift keying (M-PSK) schemes are adopted for the inner-layer subcarrier modulation and the resulting middlewares are further modulated using phase modulation (PM) or frequency modulation (FM). Assume that the CM candidate set is given by $\mathbb{K} \overset{\text{def}}{=} \{\mathscr{K}_1, \mathscr{K}_2, \cdots, \mathscr{K}_K\}$, where $\mathscr{K}_k$ denotes the k^{th} CM scheme, k=1, 2, …, K.

Consider the CM with $\mathscr{J}$ subcarriers involved in the cognitive TT&C and aerospace communication systems. Multiple users' information-data streams are multiplexed onto the i^{th} subcarrier, i=1, 2, …, $\mathscr{J}$, all of which are encoded by the non-return-to-zero-level (NRZ-L) pulse code modulation (PCM). The inner M-PSK modulated signal $s_i(t)$ of the i^{th} subcarrier can be expressed by

$$s_i(t) \overset{\text{def}}{=} \Re\left\{ g_i(t) \exp\left[\jmath\left(2\pi \mathscr{f}_i t + \frac{2\pi\,(m_i-1)}{M_i} + \phi_{i,0}\right)\right]\right\},$$
$$m_i = 1, 2, \ldots, M_i,\ \ 0 \le t \le T_i, \tag{1}$$

where M_i denotes the order of the PSK modulation applied on the i^{th} subcarrier, $\phi_{i,0}$ and $\mathscr{f}_i$ represent the initial phase and the center frequency of the i^{th} subcarrier, respectively, T_i denotes the corresponding symbol period, and $g_i(t)$ specifies the pulse waveform as given by

$$g_i(t) \overset{\text{def}}{=} \begin{cases} 1, & 0 \le t \le T_i, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

According to [45], the symbol rate of the inner PCM/M-PSK modulation is limited within the range: 7.8125 symbols/second (Sps)$\le R_s \le$4000 Sps and the subcarrier frequency $\mathscr{f}_i$ should be an integer multiple of the corresponding symbol rate. Once the inner-modulated signals, $s_i(t)$, i=1, 2, …, $\mathscr{J}$ are generated, they are further modulated onto the main carrier using FM or PM.

For the PCM/M-PSK/PM modulation, the transmitted passband signal $s_{\text{PCM/PSK/PM}}(t)$ with $\mathscr{J}$ subcarriers can be expressed by

$$s_{\text{PCM/PSK/PM}}(t) \overset{\text{def}}{=} \Re\left\{x_{\text{B}}^{\text{PM}}(t) \exp\left[\jmath\left(2\pi f_p t + \Phi_0\right)\right]\right\}, \tag{3}$$

where $x_{\text{B}}^{\text{PM}}(t)$ is the equivalent complex baseband signal as given by

$$x_{\text{B}}^{\text{PM}}(t) \overset{\text{def}}{=} A \exp\left[\jmath \mathscr{K}_{\text{PM}} \sum_{i=1}^{\mathscr{J}} s_i(t)\right], \tag{4}$$

where A, f_p, and Φ_0 denote the amplitude, the main carrier frequency, and the initial carrier phase of the PCM/M-PSK/PM signal, respectively; $\mathscr{K}_{\text{PM}}$ expresses the phase-modulation index for the outer-layer primary PM and its normal range is $0.2 \le \mathscr{K}_{\text{PM}} \le 1.4$ according to [45]. Similarly, for the PCM/M-PSK/FM modulation, the transmitted passband

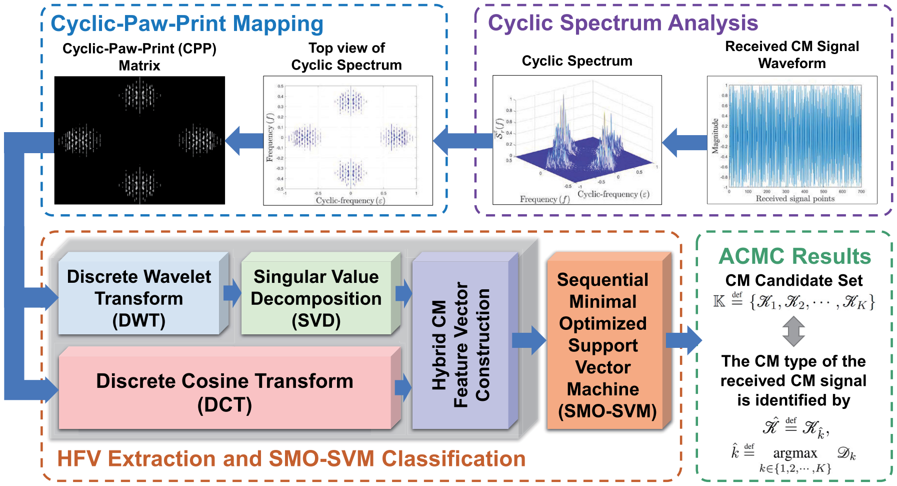

Fig. 2. The system model of our proposed new ACMC method using hybrid feature vectors of CPP.

signal $s_{\text{PCM/PSK/FM}}(t)$ with $\mathscr{J}$ subcarriers can be expressed by

$$s_{\text{PCM/PSK/FM}}(t) \overset{\text{def}}{=} \Re\{x_{\text{B}}^{\text{FM}}(t) \exp[\jmath(2\pi f_p t + \Phi_0)]\}, \tag{5}$$

where $x_{\text{B}}^{\text{FM}}(t)$ denotes the equivalent complex baseband signal as given by

$$x_{\text{B}}^{\text{FM}}(t) \overset{\text{def}}{=} A \exp\left[\jmath \mathscr{K}_{\text{FM}} \int_0^t \sum_{i=1}^{\mathscr{J}} s_i(\zeta)\, d\zeta\right], \tag{6}$$

where $\mathscr{K}_{\text{FM}}$ denotes the frequency-modulation index for the outer-layer primary FM and its normal range is $0.2 \le \mathscr{K}_{\text{FM}} \le 1.4$ according to [45].

The received CM signal $r(t)$ is distorted by the propagation channel and the channel impairment is often characterized as additive white Gaussian noise (AWGN) along with Doppler and phase noises and can thus be expressed by

$$r(t) \overset{\text{def}}{=} \Re\{\bar{x}_{\text{B}} \exp[\jmath(\varphi_{\text{d}}(t) + \varphi_{\text{p}})]\} + n(t), \tag{7}$$

where $\varphi_{\text{d}}(t)$ denotes the carrier phase evolution associated with the Doppler dynamics according to [46], φ_{p} denotes the phase noise induced by the propagation channel, which involves the initial carrier phase Φ_0 according to [46], and $n(t)$ denotes the AWGN with zero mean and variance σ^2. Note that $\bar{x}_{\text{B}}$ denotes the equivalent received complex baseband signal as given by

$$\bar{x}_{\text{B}} \overset{\text{def}}{=} \begin{cases} A \exp\left[\jmath \mathscr{K}_{\text{PM}} \sum_{i=1}^{\mathscr{J}} \tilde{s}_i(t)\right] & \text{for PCM/}M\text{-PSK/PM,} \\ A \exp\left[\jmath \mathscr{K}_{\text{FM}} \int_0^t \sum_{i=1}^{\mathscr{J}} \tilde{s}_i(\zeta)\, d\zeta\right] & \text{for PCM/}M\text{-PSK/FM,} \end{cases} \tag{8}$$

where

$$\tilde{s}_i(t) \overset{\text{def}}{=} \Re\left\{g_i(t) \exp\left\{\jmath 2\pi\left[\tilde{f}_i t + \frac{m_i - 1}{M_i}\right]\right\}\right\},$$
$$m_i = 1, 2, \ldots, M_i,\ 0 \le t \le T_i, \tag{9}$$

and $\tilde{f}_i$ denotes the i^{th} received subcarrier frequency, $\tilde{f}_i \overset{\text{def}}{=} f_i + f_{\text{d}}$, and f_{d} represents the Doppler frequency offset.

III. Our Proposed Novel ACMC Method

In an intelligent receiver, first the Doppler and carrier frequency offsets are eliminated and the phase noise is mitigated to enhance the received CM signal using high-precision blind synchronization [46]. Then our proposed new ACMC method is adopted by the cognitive-radio-based cognitive engine of payload (as illustrated by Fig. 1) to blindly identify the CM scheme. The system diagram of our proposed new ACMC approach is illustrated by Fig. 2 (highlighted as "ACMC using cyclic-paw-print"). It includes four major mechanisms, namely (*i*) *cyclic spectrum analysis*, (*ii*) *cyclic-paw-print mapping*, (*iii*) *hybrid feature vector extraction*, and (*iv*) *SMO-SVM classification*.

In our proposed ACMC scheme, the second-order cyclic spectrum analysis is applied to the aforementioned preprocessed (enhanced) CM signal to produce the corresponding three-dimensional normalized second-order cyclic spectrum. Then, the cyclic-paw-print mapping mechanism is utilized to construct the CPP matrix from the top view of the second-order cyclic spectrum. Once such a CPP is established, it is simultaneously transformed by the two-dimensional DCT and the two-dimensional DWT using *Haar wavelet*. The DCT feature is in favor here due to its outstanding *energy-compaction* property and thus it can digest the extracted CPP matrix while the DWT is invoked here to further facilitate the multi-resolution analysis in both time and frequency domains. The hybrid feature vector can be acquired from the low-frequency components of the resulted DCT matrix and the principal components of the resulted DWT matrix. Based on the hybrid feature vectors obtained from the training and test CM signal data, the CM scheme of the received signal can be identified using the SMO-SVM classifier. The four major mechanisms of our proposed novel ACMC scheme are detailed in the following subsections.

A. Cyclic Spectrum Analysis

The periodicity of second-order statistics of the CM signals can be explored for ACMC according to our previous study of AMC using cyclic-spectrum based features

[47], [48], [49]. Here, the second-order cyclic spectrum is exploited to characterize the received CM signal in the polyspectral domain. The second-order cyclic autocorrelation function of the discrete-time preprocessed CM signal sequence $r'(n)$ can be formed using the FFT (fast Fourier transform) accumulation method (FAM) [49]. According to [49], the time-smooth-based cyclic periodogram $\mathcal{S}^{\varepsilon}_{r_T}(n,f)_{\Delta t}$ for a given spectral-frequency f and a cyclic-frequency ε can be calculated as

$$\mathcal{S}^{\varepsilon}_{r_T}(n,f)_{\Delta t} \overset{\text{def}}{=} \sum_{\lambda} R_T(\lambda,f_1)\, R^*_T(\lambda,f_2)\, g'(n-\lambda), \quad (10)$$

where $g'(n)$ is a unity-area weighting function of width $\Delta t \overset{\text{def}}{=} NT_s$ seconds, N denotes the number of samples per FFT window, T_s is the sampling period, f_1 and f_2 denote the center frequencies of the filters used in the FAM, i.e., $f_1 \overset{\text{def}}{=} f+\varepsilon/2$, $f_2 \overset{\text{def}}{=} f-\varepsilon/2$, and $R_T(\lambda,f_1)$, $R_T(\lambda,f_2)$ are the complex demodulates of $r'(n)$, which can be obtained as

$$R_T(\lambda,f) \overset{\text{def}}{=} \sum_{\lambda=-N'/2}^{N'/2-1} \hat{w}(\lambda)\, r'(n-\lambda)\, e^{-\jmath 2\pi f(n-\lambda)T_s}. \quad (11)$$

Note that $\hat{w}(\lambda)$ denotes a data-tapering window of duration $T=N'T_s$ seconds and its bandwidth coincides with the frequency resolution Δf of the SCS. Thus, the cyclic autocorrelation function can be estimated from the time-smoothed cyclic periodogram $\mathcal{S}^{\varepsilon}_{r_T}(n,f)_{\Delta t}$ without bias such that

$$\mathcal{S}^{\varepsilon}_{r}(f) \overset{\text{def}}{=} \lim_{\Delta f\to 0}\lim_{\Delta t\to 0} \mathcal{S}^{\varepsilon}_{r_T}(n,f)_{\Delta t}. \quad (12)$$

The SCS $\mathcal{S}^{\varepsilon}_{r}(f)$ given by Eq. (12) is a three-dimensional spectrum with non-negative amplitudes and contains $2N+1$ cyclic-frequencies $\varepsilon=\varepsilon_p$, $p=-N, -N+1, \ldots, N$ and $N'+1$ spectral-frequencies $f=f_q$, $q=-N'/2, -N'/2+1, \ldots, N'/2$. Furthermore, the normalized SCS (NSCS) can be obtained as

$$\overline{\mathcal{S}}^{\varepsilon}_{r}(f) \overset{\text{def}}{=} \frac{\mathcal{S}^{\varepsilon}_{r}(f)}{\max\limits_{\varepsilon,f} \mathcal{S}^{\varepsilon}_{r}(f)}. \quad (13)$$

The NSCSs of the ideal (noise-free) CM signals specified in [45] and their top views (CPPs) are depicted by Fig. 3.

B. Cyclic-Paw-Print Mapping

According to Fig. 3, it is apparent that the top views of the cyclic spectra of the CM signals are concentrated in the four symmetrically distributed regions of the $\varepsilon-f$ plane, where in each region the scatter plot resembles an animal's paw print. Thus we call the top view of an NSCS a "*cyclic-paw-print* (CPP)". Form Fig. 3, it is conspicuous that the CPPs resulting from different CM signals are distinguishable. Now the CPP mapping mechanism is theoretically derived. The top view of an NSCS $\overline{\mathcal{S}}^{\varepsilon}_{r}(f)$ can be expressed by a P-by-Q matrix $\tilde{\mathcal{S}}_r \overset{\text{def}}{=} \left[\tilde{\mathcal{S}}_r(\rlap{p}, \rlap{q})\right]_{0\le \rlap{p} \le P-1, 0\le \rlap{q}\le Q-1}$ where $P=2N+1$ and $Q=N'+1$.

The entry $\tilde{\mathcal{S}}_r(\rlap{p},\rlap{q})$ in $\tilde{\mathcal{S}}_r$ is the non-negative and normalized ampulitude of $\overline{\mathcal{S}}^{\varepsilon}_{r}(f)$ at the cyclic frequency $\varepsilon_{\rlap{p}}$ and the spectral frequency $f_{\rlap{q}}$, where $\rlap{p}=p+N$ and $\rlap{q}=q+N'/2$. Let's further quantize the entries $\tilde{\mathcal{S}}_r(\rlap{p},\rlap{q})$'s into 16-bit binary integers such that

$$\tilde{\mathcal{S}}(\rlap{p},\rlap{q}) \overset{\text{def}}{=} \left\lfloor (2^{16}-1)\times \tilde{\mathcal{S}}_r(\rlap{p},\rlap{q})\right\rfloor, \quad \rlap{p}=0,1,,\ldots,P-1,\ \rlap{q}=0,1,\ldots,Q-1, \quad (14)$$

where "$\lfloor\ \rfloor$" denotes the integer rounding-down (flooring) operation. Thus, the CPP of the CM signal can be established by converting the top view of the NSCS $\overline{\mathcal{S}}^{\varepsilon}_{r}(f)$ into a $P\times Q$ grayscale "*CPP matrix*" represented by $\tilde{\mathcal{S}} \overset{\text{def}}{=} \left[\mathcal{S}(\rlap{p},\rlap{q})\right]_{1\le \rlap{p}\le P, 1\le \rlap{q}\le Q}$ according to Eqs. (13) and (14).

C. Hybrid Feature Vector Extraction

If one observes the CPPs of the CM signals illustrated by Fig. 3, the point-clouds (scatter points) are concentrated in four symmetrically-distributed regions and the corresponding CPP matrices $\tilde{\mathcal{S}}$'s are all sparse. Hence, the discrete cosine transform (DCT) and the discrete wavelet transform (DWT) based on Haar wavelet are simultaneously employed to extract the essential CM signal-features. As a result, a new hybrid feature vector can be formed thereby.

1) CM Feature Extraction Using DCT: The DCT, which decomposes the image into spectral sub-bands, has a favorable energy-compaction property and can lead to quality image compression with large compression ratios. Here the DCT is employed to represent a CPP matrix $\tilde{\mathcal{S}}$ as a sum of sinusoids of various magnitudes and frequencies and extract the low-frequency components as critical features. Without loss of generality, the two-dimensional Type-II DCT is adopted in this work along the rows of $\tilde{\mathcal{S}}$ and then along its columns. The ultimate DCT-coefficient matrix, denoted by $\tilde{D} \overset{\text{def}}{=} \{\tilde{D}(u,v)\}_{1\le \rlap{p}\le P, 1\le \rlap{q}\le Q}$, of a CPP $\tilde{\mathcal{S}}$ can be obtained by

$$\tilde{D}(u,v) \overset{\text{def}}{=} \sum_{\rlap{p}=0}^{P-1}\sum_{\rlap{q}=0}^{Q-1} \beta_u\,\beta_v\,\tilde{\mathcal{S}}(\rlap{p},\rlap{q}) \times \cos\left[\frac{\pi(2\rlap{p}+1)u}{2P}\right]\cos\left[\frac{\pi(2\rlap{q}+1)v}{2Q}\right], \quad u=0,1,\ldots,P-1,\ v=0,1,\ldots,Q-1, \quad (15)$$

where β_u and β_v denote the two corresponding multiplicative factors as given by

$$\beta_u \overset{\text{def}}{=} \begin{cases} \sqrt{\frac{1}{P}}, & \text{if } u=0,\\ \sqrt{\frac{2}{P}}, & \text{if } 1\le u\le P-1, \end{cases} \quad (16)$$

$$\beta_v \overset{\text{def}}{=} \begin{cases} \sqrt{\frac{1}{Q}}, & \text{if } v=0,\\ \sqrt{\frac{2}{Q}}, & \text{if } 1\le v\le Q-1. \end{cases} \quad (17)$$

The DCT-coefficient matrix $\tilde{D}$ given by Eq. (15) can be roughly divided into two parts: the low-frequency and high-frequency parts. The low-frequency coefficients, which gather around the upper-left corner of $\tilde{D}$, are larger in magnitude than other entries in $\tilde{D}$ and reserve most information of a CPP. Thus, the subset of these low-frequency coefficients in $\tilde{D}$ can be adopted as the essential features for ACMC.

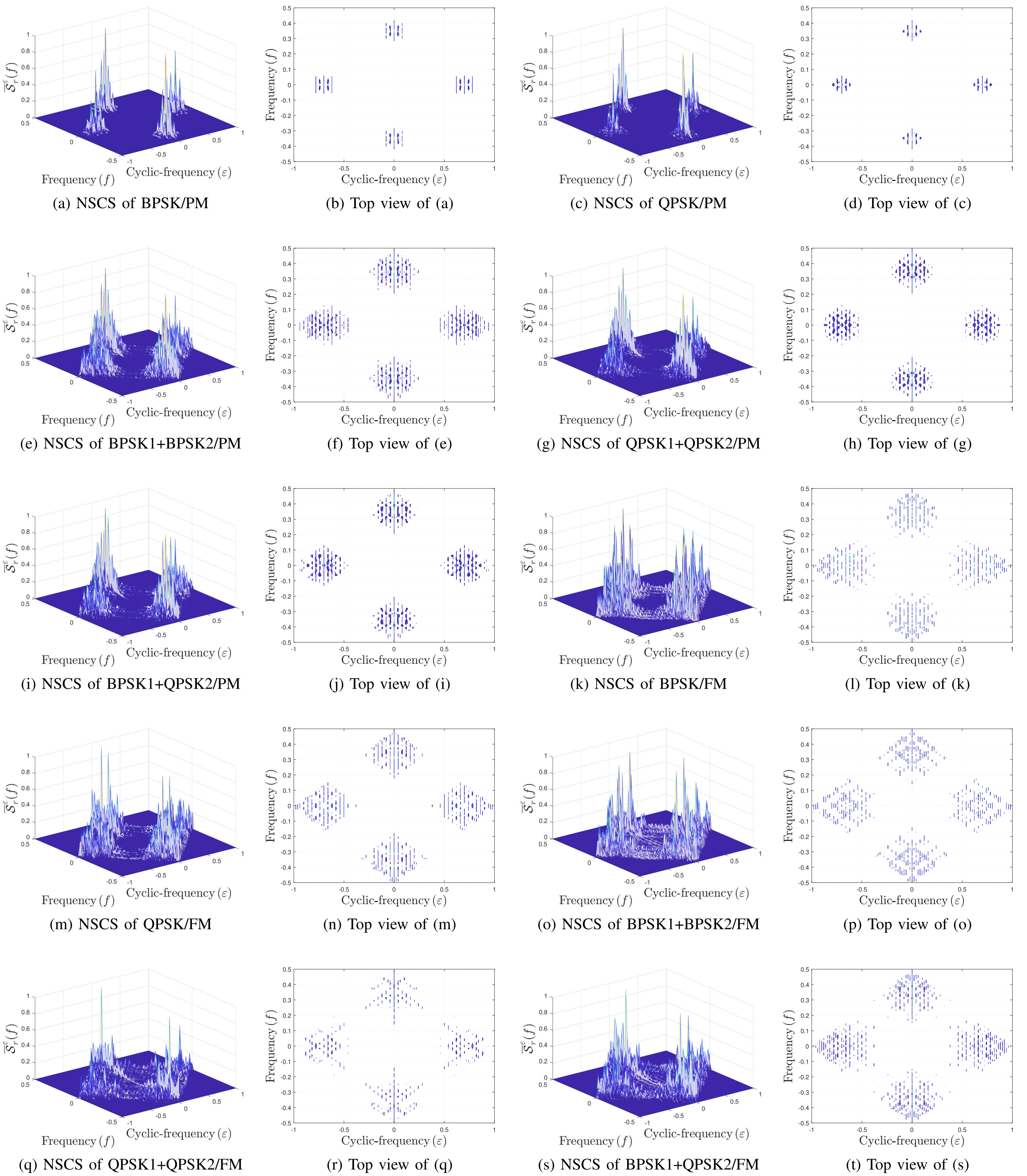

Fig. 3. The NSCSs and their top views (CPPs) of the CM signals specified in [45].

2) CM Feature Extraction Using DWT: Meanwhile, the two-dimensional discrete wavelet transform (DWT) based on Haar wavelet is adopted in this work to decompose the CPP matrix $\tilde{\mathcal{S}}$ into subbands. It should be pointed out that other commonly used wavelets, such as Daubechies wavelet with two vanishing moments (denoted by "db2" in Fig. 13) and discrete Meyer wavelet (denoted by "dmey" in Fig. 13), can also be employed here but both of them involve more wavelet coefficients than Haar wavelet. Thus, for better computationally efficiency, we propose to use Haar wavelet in this work.

The $\mathscr{N}\times\mathscr{N}$ discrete Haar wavelet transform operator is given by

$$\tilde{W}_{\mathscr{N}} \overset{\text{def}}{=} \left[\begin{array}{c}\tilde{H}_{\mathscr{N}/2}\\ \hdashline \tilde{G}_{\mathscr{N}/2}\end{array}\right] = \left[\begin{array}{c}\tilde{I}_{\mathscr{N}/2}\otimes\left[\sqrt{2}/2 \quad \sqrt{2}/2\right]\\ \hdashline \tilde{I}_{\mathscr{N}/2}\otimes\left[\sqrt{2}/2 \quad -\sqrt{2}/2\right]\end{array}\right], \tag{18}$$

where $\mathscr{N}$ is an even number and $\tilde{I}_{\mathscr{N}/2}$ denotes the $\mathscr{N}/2\times\mathscr{N}/2$ identity matrix. The $\mathscr{N}/2\times\mathscr{N}$ submatrices $\tilde{H}_{\mathscr{N}/2}$ and $\tilde{G}_{\mathscr{N}/2}$ represent the low-pass and high-pass subfilters, respectively. To perform the two-dimensional DWT of the CPP matrix $\tilde{\mathscr{S}}$, the columns and rows of $\tilde{\mathscr{S}}$ are treated separately. The "*DWT matrix*" $\tilde{B}$ is thus given by

$$\begin{aligned}\tilde{B} &\overset{\text{def}}{=} \tilde{W}_P\,\tilde{\mathscr{S}}\,\tilde{W}_Q^T = \begin{bmatrix}\tilde{H}_{P/2}\\ \tilde{G}_{P/2}\end{bmatrix}\tilde{\mathscr{S}}\begin{bmatrix}\tilde{H}_{Q/2}\\ \tilde{G}_{Q/2}\end{bmatrix}^T\\ &= \left[\begin{array}{c:c}\underbrace{\tilde{H}_{P/2}\,\tilde{\mathscr{S}}\,\tilde{H}_{Q/2}^T}_{\tilde{C}} & \underbrace{\tilde{H}_{P/2}\,\tilde{\mathscr{S}}\,\tilde{G}_{Q/2}^T}_{\tilde{E}}\\ \hdashline \underbrace{\tilde{G}_{P/2}\,\tilde{\mathscr{S}}\,\tilde{H}_{Q/2}^T}_{\tilde{F}} & \underbrace{\tilde{G}_{P/2}\,\tilde{\mathscr{S}}\,\tilde{G}_{Q/2}^T}_{\tilde{Y}}\end{array}\right] = \left[\begin{array}{c:c}\tilde{C} & \tilde{E}\\ \hdashline \tilde{F} & \tilde{Y}\end{array}\right].\end{aligned} \tag{19}$$

Consequently, the DWT matrix $\tilde{B}$ of the CPP matrix $\tilde{\mathscr{S}}$ can be decomposed into submatrices according to Eq. (19). The submatrix $\tilde{C}$ located at the upper-left corner of the DWT matrix $\tilde{B}$ represents an *approximation* of the CPP matrix $\tilde{\mathscr{S}}$, which is attained by successively averaging the columns and rows of $\tilde{\mathscr{S}}$. The remaining sparse submatrices, namely $\tilde{E}$, $\tilde{F}$, and $\tilde{Y}$, exhibit the vertical, horizontal, and diagonal differences of $\tilde{\mathscr{S}}$, respectively. Since the energy of a CPP matrix $\tilde{\mathscr{S}}$ mostly concentrates in $\tilde{C}$, we just retain $\tilde{C}$ as the concise representation of $\tilde{\mathscr{S}}$. Note that we may recursively perform the DWT of $\tilde{C}$ again and again such that

$$\tilde{C}_\ell \overset{\text{def}}{=} \begin{cases}\tilde{H}_{P/2^\ell}\,\tilde{C}_{\ell-1}\tilde{H}_{Q/2^\ell}^T, & \text{if } \ell=2,3,\ldots,\\ \tilde{H}_{P/2^\ell}\,\tilde{\mathscr{S}}\tilde{H}_{Q/2^\ell}^T, & \text{if } \ell=1,\end{cases} \tag{20}$$

where $\tilde{C}_1{=}\tilde{C}$ and ℓ is the "*decomposition-level index*". Furthermore, the singular value decomposition (SVD) can also be applied to $\tilde{C}$ for dimensionality reduction and the corresponding singular values are utilized to form the hybrid feature vector. The SVD of $\tilde{C}$ at Level ℓ is given by

$$\begin{aligned}\tilde{U}_\ell^H\,\tilde{C}_\ell\,\tilde{V}_\ell = \tilde{\Sigma}_\ell = \operatorname{diag}\left(\sigma_1,\sigma_2,\cdots,\sigma_{\vartheta_\ell}\right)\in\mathbb{R}^{P/2^\ell\times Q/2^\ell},\\ \ell=1,2,\ldots,\end{aligned} \tag{21}$$

where $\sigma_1{\ge}\sigma_2{\ge}\cdots{\ge}\sigma_{\vartheta_\ell}{\ge}0$; $\operatorname{diag}\left(\sigma_1,\sigma_2,\cdots,\sigma_{\vartheta_\ell}\right)$ denotes the diagonal matrix whose diagonal elements are σ_1, σ_2, $\ldots$, σ_{ϑ_ℓ}; $\vartheta_\ell\overset{\text{def}}{=}\min\left\{P/2^\ell,Q/2^\ell\right\}$; $\tilde{U}_\ell{\in}\mathbb{R}^{P/2^\ell\times P/2^\ell}$ and $\tilde{V}_\ell{\in}\mathbb{R}^{Q/2^\ell\times Q/2^\ell}$ are the unitary matrices whose columns are the eigenvectors of $\tilde{C}_\ell\tilde{C}_\ell^H$ and $\tilde{C}_\ell^H\tilde{C}_\ell$, respectively; $\tilde{\Sigma}_\ell$ denotes a diagonal matrix which contains all singular values of $\tilde{C}_\ell$, namely σ_κ, $\kappa{=}1$, 2, $\ldots$, ϑ_ℓ. If the rank of $\tilde{C}_\ell$ is $\gamma_{\tilde{C}_\ell}$, then $\gamma_{\tilde{C}_\ell}{\le}\vartheta_\ell$ and $\sigma_1{\ge}\sigma_2{\ge}\cdots{\ge}\sigma_{\gamma_{\tilde{C}_\ell}}{>}0$, $\sigma_{\gamma_{\tilde{C}_\ell}+1}{=}\sigma_{\gamma_{\tilde{C}_\ell}+2}{=}\cdots{=}\sigma_{\vartheta_\ell}{=}0$. In this work, the positive singular values: σ_1, σ_2, $\ldots$, $\sigma_{\gamma_{\tilde{C}_\ell}}$ are adopted to form the new hybrid feature vector of a CM signal.

3) Hybrid Feature Vector (HFV) Construction Using Both DCT and DWT: Once the DCT-coefficient matrix $\tilde{D}$ and the positive singular values of the CPP matrix $\tilde{\mathscr{S}}$ are obtained according to Section III-C.1 and III-C.2, respectively, we can form the hybrid feature vector. The sums of the squares of the elements in each row and column of $\tilde{D}$ are depicted by Fig. 4 and Fig. 5, respectively. It is obvious that the entries located at the first few entries of the first row of $\tilde{D}$ have significantly larger magnitudes than others. According to our heuristic analysis, we propose to extract the significant entries of $\tilde{D}$, $\tilde{\Sigma}_1$, and $\tilde{\Sigma}_2$ for form the new hybrid feature vector $\vec{Z}$ of a CM signal such that

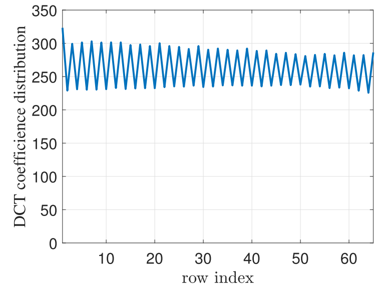

Fig. 4. The row-wise sums of the squares of the elements of $\tilde{D}$.

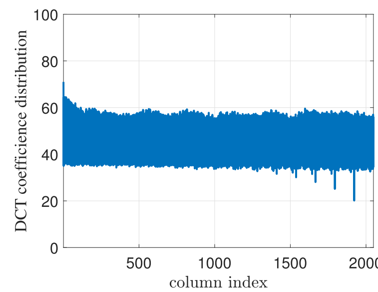

Fig. 5. The column-wise sums of the squares of the elements of $\tilde{D}$.

$$\begin{aligned}\vec{Z} \overset{\text{def}}{=} \Big[&\tilde{D}(1,1)\ \tilde{D}(1,2)\ \cdots\ \tilde{D}(1,6)\ \tilde{\Sigma}_1(1,1)\ \tilde{\Sigma}_1(2,2)\ \cdots\\ &\tilde{\Sigma}_1(6,6)\ \tilde{\Sigma}_2(1,1)\ \tilde{\Sigma}_2(2,2)\ \cdots\ \tilde{\Sigma}_2(6,6)\Big]^T,\end{aligned} \tag{22}$$

where $\tilde{D}(\mathscr{p},\mathscr{q})$, $\tilde{\Sigma}_1(\mathscr{p},\mathscr{q})$, and $\tilde{\Sigma}_2(\mathscr{p},\mathscr{q})$ represent the $(\mathscr{p},\mathscr{q})^{\text{th}}$ entries of $\tilde{D}$, $\tilde{\Sigma}_1$, and $\tilde{\Sigma}_2$, respectively. The hybrid feature vector $\vec{Z}$ given by Eq. (22) will be employed for ACMC in this work. The nonlinear SVM classifier can be built as

$$\Omega_k(\vec{Z}) + \mathscr{b}_k \underset{\substack{\vec{Z}\text{ is not the }k^{\text{th}}\\ \text{modulation scheme}}}{\overset{\substack{\vec{Z}\text{ is the }k^{\text{th}}\\ \text{modulation scheme}}}{\gtrless}} 0,\ k=1,2,\ldots,K, \tag{23}$$

where $\Omega_k(\vec{Z})$ denotes the optimal nonlinear projection function of $\vec{Z}$ and $\mathscr{b}_k$ denotes the optimal bias, both to be determined.

For the training signal (in the presence of noise) using the k^{th} CM scheme indexed by $\mathscr{K}_k{\in}\mathbb{K}$, one can build the CPPs from the corresponding NSCSs in ρ trials and construct the corresponding hybrid feature vector set, namely $\mathbb{Z}_k^{\text{training}}\overset{\text{def}}{=}\left\{\vec{Z}_1^k,\vec{Z}_2^k,\cdots,\vec{Z}_\rho^k\right\}$ where $\vec{Z}_\mu^k$ denotes the hybrid feature vector $\vec{Z}$ generated from the k^{th} CM scheme in the μ^{th} trial according to Eq. (22), $k{=}1$, 2, $\ldots$, K and $\mu{=}1$, 2, $\ldots$, ρ. Moreover, the training hybrid feature vector set for the entire CM candidate set $\mathbb{K}$ can also be established as $\mathbb{Z}^{\text{training}} \overset{\text{def}}{=} \{\mathbb{Z}_1^{\text{training}}, \mathbb{Z}_2^{\text{training}}, \cdots, \mathbb{Z}_K^{\text{training}}\}$.

D. SMO-SVM Classification

Once the hybrid feature vector of a received CM signal is built from the corresponding CPP using our proposed approach stated in Section III-C.3, the unknown CM scheme can be determined using the support vector machines where a CM type $\mathscr{K}_k$ is selected from $\mathbb{K}$ accordingly as the underlying modulation scheme of the received CM signal. Since there are K CM-scheme candidates in $\mathbb{K}$, this turns out to be a multi-classification problem consisting of K classes, which can be further reduced to K binary classification subproblems by involving K separate binary classifiers. Then the multi-class SVM can be established by K *sequential minimal optimized support vector machines* (SMO-SVMs) as depicted in Fig. 6. Each SMO-SVM (indexed by 1 to K) illustrated by Fig. 6 distinguishes between one CM candidate $\mathscr{K}_k \in \mathbb{K}$ and the rest (one-versus-all-others) and the winner-takes-all strategy is in place such that the SMO-SVM with the highest output-objective value will dominate the overall decision.

1) Training Stage: The multi-class SVM is first trained using the aforementioned hybrid feature vector set $\mathbb{Z}^{\text{training}}$ generated from the training CM signals (subject to various noise levels). For the k^{th} SMO-SVM as illustrated in Fig. 6, the k^{th} CM-scheme candidate $\mathscr{K}_k$ is distinguished from the other CM-scheme candidates in $\mathbb{K}$. For a given hybrid feature vector $\vec{Z}_\mu^{k'}$, let's define

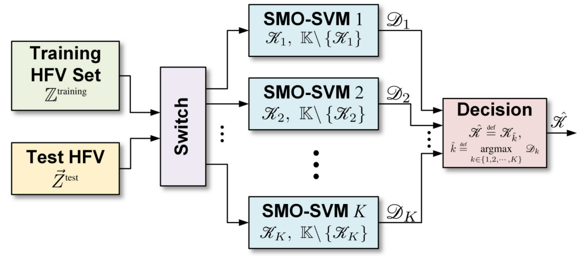

Fig. 6. Our proposed multi-class SVM (SMO-SVMs) for ACMC.

$$\delta(k,k') \overset{\text{def}}{=} \begin{cases} 1, & \text{if } k' = k, \\ -1, & \text{if } k' \neq k. \end{cases} \tag{24}$$

Thus, according to [50], the typical optimization problem of SVM is equivalent to solving as in Eq. (25), shown at the bottom of the page. Once the optimization problem formulated by Eq. (25) is solved, the maximum margin hyperplane of the k^{th} SMO-SVM can be obtained, where all corresponding optimal Lagrange multipliers satisfy the *Karush-Kuhn-Tucker* (KKT) conditions. In order to reduce the high computational complexity incurred in solving Eq. (25), the SMO technique, which is a heuristic algorithm to optimize the objective function of SVM, is then invoked here to continuously decompose the complicated quadratic programming (QR) problem into multiple subproblems, each of which involves only two independent variables, where at least one of these two variables violates the KKT conditions. Nonetheless, after iteratively

$$\begin{aligned} &\min_{\vec{\alpha}_k} \ \frac{1}{2}\sum_{\mu=1}^{\rho}\sum_{\mu'=1}^{\rho}\sum_{k'=1}^{K}\sum_{k''=1}^{K} \alpha_{k,k'}\,\alpha_{k,k''}\,\delta_{k,k'}\,\delta_{k,k''}\,\mathtt{K}\left(\vec{Z}_\mu^{k'}, \vec{Z}_{\mu'}^{k''}\right) - \sum_{k'=1}^{K}\sum_{\mu=1}^{\rho}\alpha_{k,k'} \\ &\text{s.t.} \ \sum_{k'=1}^{K}\sum_{\mu=1}^{\rho}\alpha_{k,k'}\,\delta_{k,k'} = 0 \text{ and } 0 \le \alpha_{k,k'}, \alpha_{k,k''} \le \mathscr{C} \text{ for } k = 1,2,\ldots,K, \end{aligned} \tag{25}$$

where $\vec{\alpha}_k \overset{\text{def}}{=} \left[\alpha_{k,k'}\right]_{1\le k'\le K}$ denotes the $K\times 1$ Lagrangian-multiplier vector for the k^{th} SVM, $\mathscr{C}$ specifies the positive penalty term that minimizes the number of misclassified samples, and $\mathtt{K}(\ )$ denotes a nonlinear kernel function.

$$\begin{aligned} &\min_{\alpha_{k,\iota_1},\alpha_{k,\iota_2}} \ \mathscr{Q}\left(\alpha_{k,1}, \alpha_{k,2}, \bar{\mathtt{K}}_{\iota_1\iota_1}, \bar{\mathtt{K}}_{\iota_1\iota_2}, \bar{\mathtt{K}}_{\iota_2\iota_2}\right) \\ &\quad \text{s.t.} \ \ \alpha_{k,\iota_1}\,\delta_{k,\iota_1} + \alpha_{k,\iota_2}\,\delta_{k,\iota_2} = -\sum_{k'=3}^{K}\delta_{k,k'}\,\alpha_{k,k'} = \varsigma \text{ and } 0 \le \alpha_{k,k'} \le \mathscr{C} \\ &\qquad \text{for } k' = 1,2,\ldots,K, \end{aligned} \tag{26}$$

where

$$\begin{aligned} \mathscr{Q}\left(\alpha_{k,1}, \alpha_{k,2}, \bar{\mathtt{K}}_{\iota_1\iota_1}, \bar{\mathtt{K}}_{\iota_1\iota_2}, \bar{\mathtt{K}}_{\iota_2\iota_2}\right) \overset{\text{def}}{=} &\ \frac{1}{2}\bar{\mathtt{K}}_{\iota_1\iota_1}\,\alpha_{k,\iota_1}^2 + \frac{1}{2}\bar{\mathtt{K}}_{\iota_2\iota_2}\,\alpha_{k,\iota_2}^2 + \delta_{k,\iota_1}\,\delta_{k,\iota_2}\,\bar{\mathtt{K}}_{\iota_1\iota_2}\,\alpha_{k,\iota_1}\,\alpha_{k,\iota_2} \\ &- \left(\alpha_{k,\iota_1} + \alpha_{k,\iota_2}\right) + \delta_{k,\iota_1}\,\alpha_{k,\iota_1}\sum_{k'=3}^{K}\delta_{k,k'}\,\alpha_{k,k'}\,\bar{\mathtt{K}}_{k'\iota_1} \\ &+ \delta_{k,\iota_2}\,\alpha_{k,\iota_2}\sum_{k'=3}^{K}\delta_{k,k'}\,\alpha_{k,k'}\,\bar{\mathtt{K}}_{k'\iota_2}, \end{aligned} \tag{27}$$

$\bar{\mathtt{K}}_{k'k''} \overset{\text{def}}{=} \mathtt{K}\left(\vec{Z}_\mu^{k'}, \vec{Z}_{\mu'}^{k''}\right)$, $1\le k'$, $k''\le K$, and ς is a pre-determined constant.

solving these subproblems, we can obtain the optimal values of all variables, i.e., all of them satisfy the KKT conditions finally. Without loss of generality, the corresponding two Lagrange-multipliers are assumed to be α_{k,ι_1} and α_{k,ι_2} while other Lagrange-multipliers are $\alpha_{k,k'}$, $k'{=}\iota_3,\ \iota_4,\ \ldots,\ \iota_K$ remain fixed for $1{\le}\iota_1,\ \iota_2,\ \ldots,\ \iota_K{\le}K$. Consequently, the objective function involved in each subproblem in the SMO-SVM scheme can be formulated by as in Eqs. (26) and (27) shown at the bottom of the previous page.

In each iteration of the SMO-SVM procedure, i.e., solving the subproblem given by Eqs. (26) and (27), the initial feasible solutions (obtained from the previous iteration) of α_{k,ι_1} and α_{k,ι_2} are denoted by $\alpha^{\text{pre}}_{k,\iota_1}$ and $\alpha^{\text{pre}}_{k,\iota_2}$, respectively, and the corresponding optimal solutions are denoted by $\alpha^{\text{new}}_{k,\iota_1}$ and $\alpha^{\text{new}}_{k,\iota_2}$, respectively. Subject to the constraints given by Eq. (26), the Lagrange-multiplier α_{k,ι_1} is given by

$$\alpha_{k,\iota_1} \overset{\text{def}}{=} (\varsigma - \delta_{k\iota_2}\,\alpha_{k,\iota_2})\,\delta_{k,\iota_1}, \tag{28}$$

and the objective function $\mathcal{Q}\big(\alpha_{k,1},\alpha_{k,2},\bar{\mathtt{K}}_{\iota_1\iota_1},\bar{\mathtt{K}}_{\iota_1\iota_2},\bar{\mathtt{K}}_{\iota_2\iota_2}\big)$ given by Eq. (27) can be reduced to a univariate function $\mathcal{Q}(\alpha_{k,\iota_2})$ by substituting Eq. (28) into Eq. (26). One can take the partial-derivative of $\mathcal{Q}(\alpha_{k,\iota_2})$ with respect to α_{k,ι_2} and make it equal to zero such that

$$\begin{aligned}\frac{\partial\,\mathcal{Q}(\alpha_{k,\iota_2})}{\partial\alpha_{k,\iota_2}} \overset{\text{def}}{=}& \,(\bar{\mathtt{K}}_{\iota_1\iota_1}+\bar{\mathtt{K}}_{\iota_2\iota_2}-2\,\bar{\mathtt{K}}_{\iota_1\iota_2})\,\alpha_{k,\iota_2}\\ &+\delta_{k,\iota_1}\,\delta_{k,\iota_2}-1-(\bar{\mathtt{K}}_{\iota_1\iota_1}-\bar{\mathtt{K}}_{\iota_1\iota_2})\,\varsigma\,\delta_{k,\iota_2}\\ &-(\xi_{k,\iota_1}-\xi_{k,\iota_2})\,\delta_{k,\iota_2}=0,\end{aligned} \tag{29}$$

where

$$\xi_{k,k'} \overset{\text{def}}{=} \sum_{\nu=3}^{\rho\times K}\alpha_{k,\nu}\,\delta_{k,\nu}\,\bar{\mathtt{K}}_{k'\nu},\quad k'=\iota_1,\iota_2. \tag{30}$$

Thus, the unconstrained optimal solution α_{k,ι_2} to $\min\limits_{\alpha_{k,\iota_2}}\ \mathcal{Q}(\alpha_{k,\iota_2})$ can be obtained as

$$\alpha^{\text{new,unc}}_{k,\iota_2} \overset{\text{def}}{=} \alpha^{\text{pre}}_{k,\iota_2}+\frac{\delta_{k,\iota_2}\,(\mathscr{E}_{k,\iota_1}-\mathscr{E}_{k,\iota_2})}{\psi}, \tag{31}$$

where $\psi{\overset{\text{def}}{=}}\bar{\mathtt{K}}_{\iota_1\iota_1}{+}\bar{\mathtt{K}}_{\iota_2\iota_2}{-}2\,\bar{\mathtt{K}}_{\iota_1\iota_2}$,

$$\begin{aligned}\mathscr{E}_{k,k'} &\overset{\text{def}}{=} \mathscr{g}\left(\vec{Z}^k_{\mu},\vec{Z}^{k'}_{\mu'}\right)-\delta_{k,k'}\\ &=\left[\sum_{\nu=1}^{\rho\times K}\alpha_{k,\nu}\,\delta_{k,\nu}\,\mathtt{K}\left(\vec{Z}^k_{\mu},\vec{Z}^{k'}_{\mu'}\right)+b_k\right]-\delta_{k,k'},\\ &\quad k'=\iota_1,\iota_2,\end{aligned} \tag{32}$$

$$\mathscr{g}\left(\vec{Z}^k_{\mu},\vec{Z}^{k'}_{\mu'}\right) \overset{\text{def}}{=} \sum_{\nu=1}^{\rho\times K}\alpha_{k,\nu}\,\delta_{k,\nu}\,\mathtt{K}\left(\vec{Z}^k_{\mu},\vec{Z}^{k'}_{\mu'}\right)+b_{k,k'}, \tag{33}$$

and $b_{k,k'}$ denotes the bias associated with the k'^{th} Lagrange-multiplier of the k^{th} SMO-SVM. According to the constraints given by Eq. (26), the optimal solution to Eq. (26) can be updated as

$$\alpha^{\text{new}}_{k,\iota_2} \overset{\text{def}}{=} \begin{cases}\mathtt{H} & \text{if } \alpha^{\text{new,unc}}_{k,\iota_2}>\mathtt{H},\\ \alpha^{\text{new,unc}}_{k,\iota_2} & \text{if } \mathtt{L}\le\alpha^{\text{new,unc}}_{k,\iota_2}\le\mathtt{H},\\ \mathtt{L} & \text{if } \alpha^{\text{new,unc}}_{k,\iota_2}<\mathtt{L},\end{cases} \tag{34}$$

$$\alpha^{\text{new}}_{k,\iota_1} \overset{\text{def}}{=} \alpha^{\text{pre}}_{k,\iota_1}+\delta_{k,\iota_1}\,\delta_{k,\iota_2}\left(\alpha^{\text{pre}}_{k,\iota_2}-\alpha^{\text{new}}_{k,\iota_2}\right), \tag{35}$$

where $\mathtt{H}$ and $\mathtt{L}$ denote the upper- and lower-bounds of the block constraint ($0{\le}\alpha_{k,k'}{\le}\mathscr{C}$ for $k'{=}1,\ 2,\ \ldots,\ K$) given by Eq. (26), respectively, such that

$$\mathtt{H} \overset{\text{def}}{=} \begin{cases}\min\left(\mathscr{C},\mathscr{C}+\alpha^{\text{pre}}_{k,\iota_2}-\alpha^{\text{pre}}_{k,\iota_1}\right) & \text{if } \delta_{k,\iota_1}\neq\delta_{k,\iota_2},\\ \min\left(\mathscr{C},\alpha^{\text{pre}}_{k,\iota_2}+\alpha^{\text{pre}}_{k,\iota_1}\right) & \text{if } \delta_{k,\iota_1}=\delta_{k,\iota_2},\end{cases} \tag{36}$$

and

$$\mathtt{L} \overset{\text{def}}{=} \begin{cases}\max\left(0,\alpha^{\text{pre}}_{k,\iota_2}-\alpha^{\text{pre}}_{k,\iota_1}\right) & \text{if } \delta_{k,\iota_1}\neq\delta_{k,\iota_2},\\ \max\left(0,\alpha^{\text{pre}}_{k,\iota_2}+\alpha^{\text{pre}}_{k,\iota_1}-\mathscr{C}\right) & \text{if } \delta_{k,\iota_1}=\delta_{k,\iota_2}.\end{cases} \tag{37}$$

Once the optimal Lagrange-multipliers $\alpha^{\text{new}}_{k,\iota_1}$ and $\alpha^{\text{new}}_{k,\iota_2}$ are attained using Eqs. (34) and (35), the corresponding (new) optimal biases $b^{\text{new}}_{k,\iota_1}$ and $b^{\text{new}}_{k,\iota_2}$ can be calculated as

$$\begin{aligned}b^{\text{new}}_{k,k'} \overset{\text{def}}{=}& -\mathscr{E}_{k,k'}-\delta_{k,\iota_1}\,\bar{\mathtt{K}}_{\iota_1k'}\left(\alpha^{\text{new}}_{k,\iota_1}-\alpha^{\text{pre}}_{k,\iota_1}\right)\\ &-\delta_{k,\iota_2}\,\bar{\mathtt{K}}_{\iota_2k'}\left(\alpha^{\text{new}}_{k,\iota_2}-\alpha^{\text{pre}}_{k,\iota_2}\right)+b^{\text{pre}}_{k,k'},\quad k'=\iota_1,\iota_2,\end{aligned} \tag{38}$$

where $b^{\text{pre}}_{k,k'}$, $k'{=}\iota_1,\ \iota_2$ denote the biases corresponding to the two Lagrange-multipliers $\alpha^{\text{pre}}_{k,\iota_1}$ and $\alpha^{\text{pre}}_{k,\iota_2}$ obtained from the previous iteration. Consequently, the optimal bias $\mathscr{b}^{\text{new}}_k$ (the solution to $\mathscr{b}_k$ involved in Eq. (23)) resulting from the current iteration can be obtained as

$$\mathscr{b}^{\text{new}}_k \overset{\text{def}}{=} \frac{1}{2}\left(b^{\text{new}}_{k,\iota_1}+b^{\text{new}}_{k,\iota_2}\right). \tag{39}$$

Now we can write

$$\Omega^{\text{new}}_k(\vec{Z}) \overset{\text{def}}{=} \sum_{\mu=1}^{\rho}\sum_{k'=1}^{K}\alpha^{\text{new}}_{k,k'}\,\delta_{k,k'}\,\mathtt{K}\left(\vec{Z},\vec{Z}^{k'}_{\mu}\right), \tag{40}$$

where $\Omega^{\text{new}}_k(\vec{Z})$ denotes the estimate of $\Omega_k(\vec{Z})$ defined by Eq. (23) resulting from the current iteration.

The optimal values of α_{k,ι_1} and α_{k,ι_2}, both of which satisfy the KKT conditions[1] corresponding to Eq. (25), can be determined according to Eqs. (34) and (35). Therefore, by iteratively solving the subproblems involving different pairs of Lagrange-multipliers $(\alpha_{k,\iota 3},\alpha_{k,\iota 4})$, $(\alpha_{k,\iota 5},\alpha_{k,\iota 6})$, ... back and forth until convergence, one can obtain the optimal Lagrange-multiplier vector $\vec{\alpha}^{\text{new}}_k{\overset{\text{def}}{=}}\left[\alpha^{\text{new}}_{k,1}\ \alpha^{\text{new}}_{k,2}\ \cdots\ \alpha^{\text{new}}_{k,K}\right]^T$ and the optimal bias $\mathscr{b}^{\text{new}}_k$ for the k^{th} SMO-SVM. Consequently, all K SMO-SVMs can be trained and the optimal nonlinear projection function set $\left\{\Omega^{\text{opt}}_1(\vec{Z}),\Omega^{\text{opt}}_2(\vec{Z}),\cdots,\Omega^{\text{opt}}_K(\vec{Z})\right\}$ and the optimal bias set $\left\{\mathscr{b}^{\text{opt}}_1,\mathscr{b}^{\text{opt}}_2,\cdots,\mathscr{b}^{\text{opt}}_K\right\}$ can be obtained[2] according to Eqs. (39) and (40).

2) Testing Stage: During the test, the hybrid feature vector $\vec{Z}^{\text{test}}$ of a received CM signal according to Eq. (22) will treated as the input to the K trained SMO-SVMs for K

[1]The KKT conditions of Eq. (25) can be found in [50].

[2]Note that one can write $\Omega^{\text{opt}}_k(\vec{Z}){=}\Omega^{\text{new}}_k(\vec{Z})$ and $\mathscr{b}^{\text{opt}}_k{=}\mathscr{b}^{\text{new}}_k$ for all k after the completion of the last iteration.

one-v.s.-the-rest binary classifications. For k^{th} SMO-SVM, the local decision metric $\mathscr{D}_k$ is given by

$$\mathscr{D}_k \stackrel{\text{def}}{=} \sum_{\mu=1}^{\rho} \sum_{k'=1}^{K} \alpha_{k,k'}^{\text{opt}}\, \delta_{k,k'}\, \mathrm{K}\left(\vec{Z}^{\text{test}}, \vec{Z}_{\mu}^{k'}\right) + b_k^{\text{opt}}. \tag{41}$$

Hence, the scheme of a received CM signal can be ultimately identified as

$$\hat{\mathscr{K}} \stackrel{\text{def}}{=} \mathscr{K}_{\hat{k}}, \tag{42}$$

where

$$\hat{k} \stackrel{\text{def}}{=} \underset{k\in\{1,2,\cdots,K\}}{\text{argmax}} \ \mathscr{D}_k \tag{43}$$

and $\mathscr{K}_{\hat{k}} \in \mathbb{K}$.

E. Our Proposed Novel ACMC Algorithm

In summary, the details of our proposed novel ACMC approach using the hybrid feature vectors of CPP can be manifested as Algorithm 1.

Algorithm 1 Our Proposed Novel ACMC Algorithm

Input: the CM-scheme candidate set $\mathbb{K}$ and the discrete-time received signal sequence $r'(n)$, n=1, 2, ..., $\mathscr{N}$;

Output: the CM scheme $\hat{\mathscr{K}}$;

1: Calculate the three-dimensional NSCS $\overline{\mathscr{S}}_r^{\varepsilon}(f)$ of the received CM signal according to Eqs. (10)-(13);

2: Transform the top view of the three-dimensional NSCS $\overline{\mathscr{S}}_r^{\varepsilon}(f)$ into the proposed CPP, which is a gray-scale image matrix $\tilde{\mathscr{S}}$ according to Section III-B;

3: Apply the Type-II DCT to the CPP $\tilde{\mathscr{S}}$ and form the DCT-coefficient matrix $\tilde{D}$ according to Eqs. (15)-(17);

4: Decompose the CPP $\tilde{\mathscr{S}}$ by the two-level two-dimensional DWT to obtain $\tilde{C}_\ell$, ℓ=1, 2 according to Eqs. (18) and (20). Then apply the SVD to $\tilde{C}_\ell$'s to obtain the corresponding singular values according to Eq. (21);

5: Construct the hybrid feature vector $\vec{Z}$ using the DCT-coefficient matrix $\tilde{D}$ and the singular-value matrices $\tilde{\Sigma}_1$ and $\tilde{\Sigma}_2$ according to Eq. (22);

6: **For training only:** establish a hybrid feature vector set $\mathbb{Z}^{\text{training}}$ from all training CM signals for the entire CM-scheme candidate set $\mathbb{K}$;

7: **For training only:** determine the optimal multipliers $\vec{\alpha}_k^{\text{opt}}$ and the optimal bias b_k^{opt} for all K SMO-SVMs (k=1, 2, ..., K) using the training hybrid feature vector set $\mathbb{Z}^{\text{training}}$ according to Section III-D.1;

8: **For test only:** input the test hybrid feature vector $\vec{Z}=\vec{Z}^{\text{test}}$ to the K trained SMO-SVMs and enumerate all local decision metrics $\mathscr{D}_k$, k=1, 2, ..., K using Eq. (41);

9: **For test only:** determine the CM scheme $\hat{\mathscr{K}}$ according to Eqs. (42) and (43).

IV. Simulation and Experiment

In this section, the performance of our proposed novel ACMC approach is first evaluated via Monte-Carlo simulations in terms of *probability of correct classification* (P_{cc}) and the *correct recognition percentage* (CRP) over the entire CM-scheme candidate set $\mathbb{K}$ versus the average signal-to-noise ratio (SNR). The definitions of probability of correct classification and correct recognition percentage can be found in [47], [48], and [49]. The *reliability* and *discriminability* of our proposed new hybrid feature vectors are also justified by experiments. Besides, the parameter selection involved in our proposed new ACMC method is investigated, and the performances of our proposed new ACMC method for the CM signals (generated by computer) in the presence of timing-, phase-, and frequency-synchronization errors are also evaluated to test its robustness. Moreover, the computational complexity of our proposed novel ACMC approach is studied. Finally, the hardware platform involving a vector signal generator, a wideband digital receiver, and a signal (spectrum) analyzer is established for real CM signal generation and acquisition while the real CM signal data are obtained to assess the performance of our proposed new ACMC technique.

TABLE I
Modulation Parameters of CM Signals

Parameter	Specification
Radio Frequency	8 GHz
Intermediate Frequency (IF)	70 MHz
Subcarrier 1's Frequency	8 kHz
Subcarrier 2's Frequency	16 kHz
Symbol Rate	4 ksps
Sampling Rate	200 MHz
Roll-off Factor	0.35
Transmission Channel	AWGN
Signal-to-Noise-Ratio (SNR)	$[-20\,\text{dB}, 20\,\text{dB}]$

The CM-scheme candidate set $\mathbb{K}$ which includes ten CM schemes involving one or two subcarriers, namely PCM/BPSK/PM, PCM/QPSK/PM, PCM/BPSK/FM, PCM/(BPSK1+BPSK2)/PM, PCM/(QPSK1+QPSK2)/PM, PCM/(BPSK1+QPSK2)/PM, PCM/(BPSK1+BPSK2)/FM, PCM/(QPSK1+QPSK2)/FM, PCM/(BPSK1+QPSK2)/FM, and PCM/QPSK/FM, is adopted here. The modulation parameters of the CM signals in $\mathbb{K}$ are given by Table I. For the CM signals adopted for our simulations, BPSK (binary phase-shift keying) or QPSK (quadrature phase-shift keying) is employed for the inner-layer modulation. In the two-subcarrier scenario, the linear combination of these two modulation schemes is adopted to realize the residual modulations of the subcarriers. Then, the subcarriers are further modulated onto the primary carrier using PM (phase modulation) or FM (frequency modulation). Assume that the CM signals are transmitted through an additive white Gaussian noise (AWGN) channel and received by an intelligent receiver. Our proposed new ACMC method is employed to identify the CM scheme of the received signal.

The modulation index $\mathscr{K}_{\text{PM}}$ or $\mathscr{K}_{\text{FM}}$ of the primary modulation scheme (PM or FM) is set to be 1.2 here. The FFT window size for generating the SCS in the FAM is set to be N=64, and the size of the CPP $\tilde{\mathscr{S}}$ is fixed to 65×2049. For each CM scheme in $\mathbb{K}$, we generate ρ=350 training noise-free

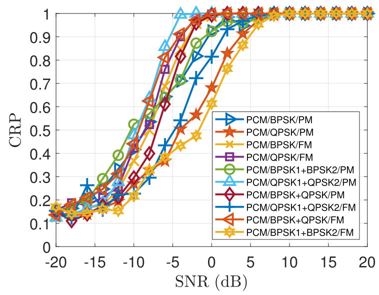

Fig. 7. Correct recognition percentages versus SNR for the given CM-scheme candidate set $\mathbb{K}$ in the presence of an AWGN channel.

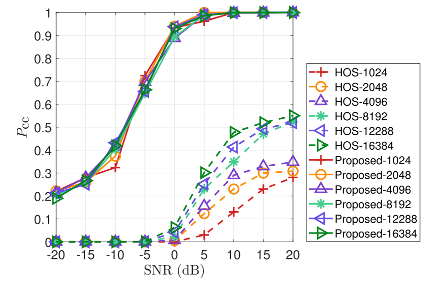

Fig. 8. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique and the existing HOS-based ACMC scheme proposed in [32] in the presence of an AWGN channel. Different sample sizes (1024, 2048, 4096, 8192, 12288, and 16384) are investigated here.

CM signals and thus can obtain 350 training hybrid feature vectors. The AWGN channel is considered and the SNR ranges from -20 to 20 dB. In each trial, 2,048 samples (symbols) are randomly generated by computer for every individual CM scheme. The ratio between the training sample size and testing sample size is set to 4:1, where the training set consists of eight hundred hybrid feature vectors and the testing set contains two hundred hybrid feature vectors, while five hundred Monte Carlo trials (except Fig. 7 where a thousand Monte Carlo trials are taken instead) are carried out for each SNR condition.

A. Performance of Our Proposed New ACMC Scheme

The CRPs for all CM-scheme candidates in $\mathbb{K}$ subject to different SNRs are delineated by Fig. 7. It is apparent from Fig. 7 that the correct recognition percentages resulting from our proposed new ACMC scheme over eight CM-scheme candidates in $\mathbb{K}$ can reach up to almost 100% when SNR≥ 5 dB. For recognizing the PCM/BPSK1+BPSK2/FM and PCM/QPSK/PM signals, the CRPs resulting from our proposed new ACMC scheme can also reach up to around 90% when the SNR is no less than 5 dB and reach up to 100% when the SNR approaches 8 dB.

The probabilities of correct classification P_{cc}'s for the aforementioned ten CM-scheme candidates resulting from our proposed new ACMC technique and the existing HOS-based (high-order-statistics-based) ACMC method proposed in [32] in the presence of an AWGN channel are depicted by Fig. 8. It is well known that the HOS features adopted in [32] are very sensitive to the sample size (received signal length). Hence, we investigate the P_{cc}'s resulting from the aforementioned two ACMC methods with respect to various sample sizes (1024, 2048, 4096, 8192, 12288, and 16384). It is obvious from Fig. 8 that the P_{cc}'s resulting from our proposed new ACMC method (denoted by "Proposed") all can converge to 1 when SNR≥ 5 dB. However, under the same SNR conditions, the P_{cc}'s resulting from the existing HOS-based ACMC method proposed in [32] (denoted by "HOS") can reach up to about 50% at best subject to the largest sample size (16384). Besides, the P_{cc}'s resulting from our proposed new ACMC method remain almost the same with respect to different sample sizes for all SNRs, which means that our proposed new ACMC method is quite insensitive to the sample size.

The existing HOS-based ACMC method proposed in [32] is a highly data-dependent approach, which requires an enormous number of signal samples to have the relatively small variances of the HOS features (estimates). To achieve a satisfactory performance, the computational complexity and processing delay (for waiting to collect many many data samples) incurred by the existing HOS-based ACMC method are both unbearably large. However, the computational resources and processing capabilities of the intelligent TT&C and aerospace communication payload carried by a space vehicle are strictly limited, which can hardly facilitate such cumbersome data processing to produce reliable HOS features in real-time. Furthermore, it should be noticed that the recognition decision of the existing HOS-based ACMC method is made by a simple thresholding mechanism. The inappropriate threshold set in the existing HOS-based ACMC method would lead to poor recognition performance. As a result, the existing HOS-based ACMC method would not be promising in practice. On the contrary, according to Fig. 8, our proposed novel CPP-based ACMC method only requires 2048 or fewer signal samples to lead to the promising recognition performance and can be conveniently employed in the realistic resource-constrained application scenarios.

Meanwhile, the performance of our proposed new ACMC method is also compared with those of the DL-based ACMC methods in Fig. 9. Note that the CPPs are extracted from the received CM signals using our proposed new ACMC method and then the CPPs are utilized as the input features of various DL networks, including ResNet in [40], MobileNet V2 in [51], and ShuffleNet in [51]. According to Fig. 9, the P_{cc} resulting from the aforementioned DL-based ACMC methods hardly can reach up to 100% in all SNR conditions. On the other hand, the P_{cc} resulting from our proposed new ACMC method (denoted by "Proposed") can reach up to around 80% when the SNR is as low as 0 dB and reach up to 100% when SNR≥ 10 dB.

B. Reliability and Discriminability of Our Proposed Novel Hybrid Features

To verify the reliability of our proposed new DWT-DCT hybrid scheme, the DCT coefficients and DWT coefficients stated in Section III-C are employed here to construct the feature vectors for SMO-SVM classification. The P_{cc}'s for the entire CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique using DCT feature vectors,

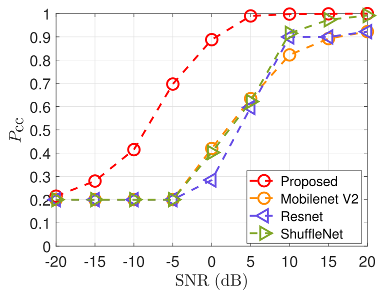

Fig. 9. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique and the existing DL-based ACMC schemes in the presence of an AWGN channel.

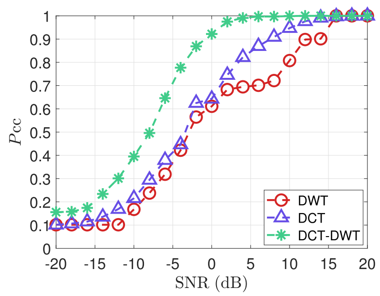

Fig. 10. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique using DCT feature vectors, DWT feature vectors, and DCT-DWT hybrid feature vectors in the presence of an AWGN channel.

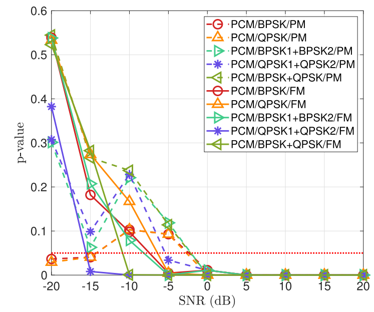

Fig. 11. The average p-values of the hybrid feature vectors corresponding to the given CM-schemes and other candidates in $\mathbb{K}$ with respect to SNR in the presence of an AWGN channel.

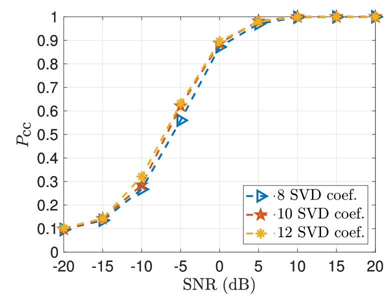

Fig. 12. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from the ten-fold cross-validation of our proposed new ACMC technique with different numbers of selected significant SVD coefficients in the presence of an AWGN channel.

DWT feature vectors, and DCT-DWT hybrid feature vectors in the presence of an AWGN channel are delineated by Fig. 10, where the simulation parameters remain unchanged.

It is conspicuous from Fig. 10 that by applying our proposed new ACMC technique using different types of feature vectors, the P_{cc}'s resulting from DCT-DWT hybrid feature vectors (denoted by "DCT-DWT") are superior to those resulting from DCT feature vectors and DWT feature vectors (denoted by "DCT" and "DWT", respectively) for all SNRs. The P_{cc}'s resulting from our proposed new ACMC scheme using DCT-DWT hybrid feature vectors can reach up to 95% even when the SNR is as low as $0\,\text{dB}$ and reach up to 100% when $\text{SNR}\geq 10\,\text{dB}$. However, the P_{cc}'s resulting from DCT feature vectors and DWT feature vectors can only lead to about 65% for $\text{SNR}=0\,\text{dB}$ and approach 100% only when $\text{SNR}\geq 16\,\text{dB}$. Consequently, by combining both DCT and DWT coefficients to form the hybrid feature vectors, one can greatly improve the recognition performance. Therefore, the reliability of our proposed new DCT-DWT hybrid feature vectors has been justified.

Meanwhile, the discriminability of our proposed new hybrid feature vectors extracted from the CPPs corresponding to the CM-schemes in $\mathbb{K}$ is investigated here using the p-ANOVA analysis according to [52], where the p-value in "*analysis of variance* (ANOVA)" determines if the difference between two group means is statistically significant. The p-valve of each pair of classes (CM-schemes) in $\mathbb{K}$ is first evaluated with same simulation settings. Then, for each CM-scheme candidate in $\mathbb{K}$, the p-values corresponding to its hybrid feature vectors and the hybrid feature vectors of all other CM-scheme candidates in $\mathbb{K}$ are averaged for every given SNR and depicted by Figure 11.

According to Fig. 11, the average p-values of the hybrid feature vectors corresponding to each pair of CM-scheme candidates in $\mathbb{K}$ are all below the threshold 0.05 (denoted by the red dash line in Fig. 11) when $\text{SNR}\geq -2\,\text{dB}$ and converge to 0 when $\text{SNR}\geq 5\,\text{dB}$. Consequently, our proposed new ACMC technique can extract the significantly distinguishable hybrid feature vectors from the constructed CPPs for ACMC. The discriminability of our proposed new hybrid feature vectors has been justified as well.

C. Parameter Selection of Our Proposed New ACMC Scheme

According to Section III-C, Haar wavelet is adopted to decomposed the CPP matrix, and some of the singular values (SVD coefficients) of the corresponding two-dimensional DWT matrix are employed to form the ultimate hybrid feature vector. Here, how to select an appropriate number of singular values and a proper wavelet are investigated using the k-fold cross-validation [53] where k=10.

For the diagonal matrices $\tilde{\Sigma}_1$ and $\tilde{\Sigma}_2$ as given by Eq. (21), the number of significant singular values in each matrix selected to generate the hybrid feature vector is set to 4, 5, and 6, respectively, which makes the number of SVD coefficients involved in the corresponding hybrid feature vector equivalent to 8, 10, and 12. The ten-fold cross-validation is carried out for each of the above-stated numbers of selected significant singular values, and thus the corresponding P_{cc}'s are depicted by Fig. 12.

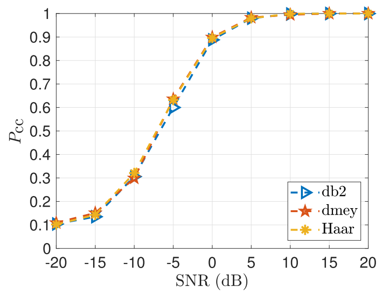

Fig. 13. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique using db2 wavelet, dmey wavelet, and Haar wavelet in the presence of an AWGN channel.

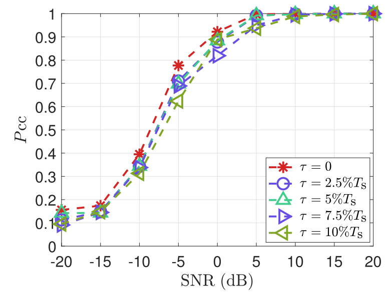

Fig. 14. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique subject to different residual timing-errors in the presence of an AWGN channel.

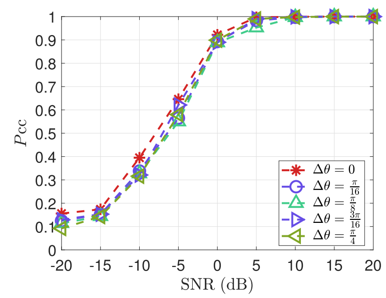

Fig. 15. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique subject to different residual phase-offsets in the presence of an AWGN channel.

It is obvious from Fig. 12 that the appropriate number of selected significant SVD coefficients for all SNRs is 12.

Then, the candidate wavelets adopted to generate hybrid feature vectors in our proposed new ACMC approach may be db2 wavelet, dmey wavelet, and Haar wavelet. The ten-fold cross-validation is also applied for each of the above-mentioned wavelets, and the corresponding P_{cc}'s are depicted by Fig. 13.

According to Fig. 13, by use of our proposed new ACMC method, the P_{cc}'s resulting from these three wavelets are almost the same. However, as Haar wavelet only consists of two coefficients, the associated excellent computational complexity is quite appealing and we propose to adopt it thereby.

D. Robustness of Our Proposed New ACMC Approach

The robustness of our proposed new ACMC technique is also investigated by evaluating its performances in the presence of timing errors, phase offsets, and frequency offsets. Assume that imperfect timing-synchronization occurs at the CM receiver. Hence the received CM signal would endure the residual timing error. Usually, the maximum timing-error that a typical timing synchronizer can handle is approximately 10% of a symbol period T_S [54]. Therefore, the residual timing error τ after timing synchronization is considered to be 2.5%, 5%, 7.5%, and 10% (the worst case) of T_S. Other simulation settings remain unchanged as before. In the presence of such residual timing errors, the corresponding probabilities of correct classification, P_{cc}'s, for the aforementioned ten CM-scheme candidates resulting from our proposed new ACMC technique are evaluated and compared to the P_{cc}'s resulting from our proposed new ACMC technique subject to perfect timing synchronization (denoted by "τ=0") in Fig. 14. It is obvious from Fig. 14 that the P_{cc}'s resulting from our proposed new ACMC technique almost stay the same for all kinds of residual timing errors in all SNR conditions. Hence, our proposed new ACMC technique is insensitive to timing errors thereby.

Furthermore, the impact of phase offset on the performance of our proposed new ACMC technique is also investigated here. Assume that imperfect phase-synchronization occurs at the CM receiver. Since the "S-curve" of a phase detector in the carrier synchronization loop adopted by the CM receiver in [55] can be modeled as a linear function only within $[-\pi/4, \pi/4]$, the residual phase-offset $\Delta\theta$ is considered to be $\pi/16$, $\pi/8$, $3\pi/16$, and $\pi/4$ (the worst case). Other simulation settings remain the same as before. The probabilities of correct classification, P_{cc}'s, for the aforementioned ten CM-scheme candidates are delineated by Fig. 15. According to Fig. 15, the P_{cc}'s resulting from our proposed new ACMC technique subject to perfect phase synchronization (denoted by "$\Delta\theta$=0") are very close to those subject to non-zero phase-offsets for all SNRs. Hence, our proposed new ACMC approach is also robust against phase-offsets.

Finally, the effect of carrier-frequency offset on the performance of our proposed new ACMC technique is investigated here. The typical scenario of a satellite telemetry transmission system using CM involves a 10 MHz bandwidth and an initial frequency error up to 2 MHz [56], [57]. Assume that imperfect frequency-synchronization occurs at the CM receiver [55]. To emulate the IF (intermediate frequency) sampling in a real CM receiver, the primary carrier frequency f_c is set to be the IF 70 MHz while the residual frequency-offset Δf is considered to be $0.025 f_c$, $0.05 f_c$, $0.075 f_c$, and $0.1 f_c$. Other simulation settings remain the same as before. The probabilities of correct classification, P_{cc}'s, for the aforementioned ten CM-scheme candidates subject to various frequency-offsets are depicted by Fig. 16. In comparison with the P_{cc}'s subject to perfect frequency-synchronization (denoted by "$\Delta f = 0$"), those subject to non-zero frequency-offsets are slightly worse. For a given SNR, the larger the frequency-offset Δf, the smaller the P_{cc} resulting from our proposed new ACMC technique according to Fig. 16. In the worst case (Δf=$0.1 f_c$),

TABLE II
THE COMPUTATIONAL COMPLEXITIES OF OUR PROPOSED NEW ACMC TECHNIQUE AND THE EXISTING DL-BASED ACMC SCHEMES

	Mobilenet V2	ResNet	ShuffleNet	Proposed
Memory	2970 kB	3472 kB	1650 kB	51 kB
Trainable parameters	2240 kB	2352 kB	1260 kB	20 kB
Training time	12 sec	16 sec	10 sec	31 sec
Testing time	0.600 ms	0.800 ms	0.450 ms	0.056 ms

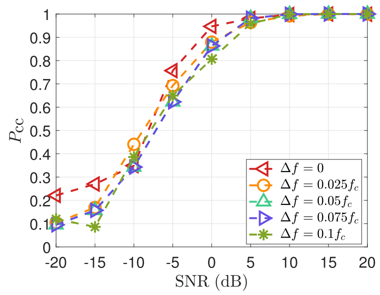

Fig. 16. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from our proposed new ACMC technique subject to different frequency-offsets in the presence of an AWGN channel.

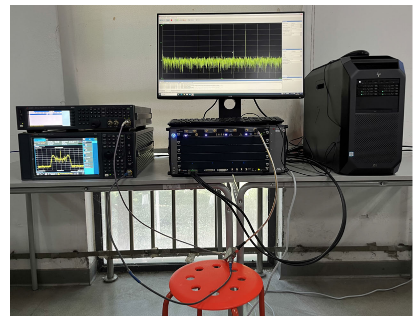

Fig. 17. The hardware platform for CM signal generation and acquisition in our laboratory.

the corresponding P_{cc} can still reach up to 80% when SNR$=0$ dB and reach up to 100% when the SNR approaches 10 dB. As a result, our proposed new ACMC approach can maintain a relatively consistent performance against various frequency-offsets.

E. Computational-Complexity Analysis

The computational complexity of our proposed new ACMC technique stated in Section III is theoretically analyzed here, which involves three parts, including the CPP construction based on the SCS anaylsis, the hybrid feature vector extraction using DCT and DWT, and the hybrid feature vector classification using multiclass SVM. Since the training CM signals are noise-free ideal signals, the training feature datasets can always be acquired and stored in advance. Therefore, we only need to consider the computational complexity of the hybrid feature vector extraction for test signals and the corresponding classifications. Meanwhile, the existing ACMC methods can be divided into two major categories: (*i*) the statistics-based approach and (*ii*) the DL-based approach, which have different computational-complexity measures.

Here, the computational complexity of our proposed new ACMC method is first theoretically analyzed and compared to that of the existing HOS-based ACMC scheme proposed in [32]. Assume that N received CM signal samples are collected for ACMC. The computational complexity of our proposed new ACMC method for one trial is given by $\mathcal{O}\left(N^2\right)+\mathcal{O}\left(N\log(N)\right)+2\,\mathcal{O}(N)+\mathcal{O}(1)\approx\mathcal{O}\left(N^2\right)$ in the worst case. On the contrary, the computational complexity of the existing HOS-based ACMC scheme for one trial is given by $2\,\mathcal{O}\left(N^4\right)+\mathcal{O}\left(N^2\right)+\mathcal{O}(1)\approx\mathcal{O}\left(N^4\right)$, which is much larger than that of our proposed new ACMC method.

On the other hand, for the DL-based ACMC approach, the computational resource and time consumption are considered as the metrics to evaluate the corresponding computational complexity. In our simulations, all ACMC methods in comparison are evaluated using the laptop equipped with an Intel Core i7-10870H CPU, 32 GB RAM, and an NVIDIA RTX 2060 GPU. The computational resource and time consumption of our proposed new ACMC technique and the existing DL-based ACMC schemes are summarized by Table II. According to Table II, our proposed new ACMC technique requires the fewest trainable parameters (in terms of 20 kB[3]) among all ACMC methods in comparison. Meanwhile, our proposed new ACMC technique requires the longest training time. However, during the testing stage, the DL-based ACMC schemes take about ten times as long as our proposed new ACMC technique. Consequently, our proposed new ACMC technique is highly memory- and computationally-efficient in comparison with the existing DL-based ACMC schemes.

F. Performance Evaluation of Our Proposed New ACMC Scheme Using Real CM Signals

At last, we test our proposed novel ACMC technique using the real CM signal data. The hardware platform for CM signal generation and acquisition is constructed in our laboratory and the photo is demonstrated by Fig. 17.

The hardware platform as shown in Fig. 17 consists of a vector signal generator (VSG), a wideband digital receiver (WDR), a workstation, and a signal analyzer (SA), which are connected by cables. The instruments and their corresponding functions are listed in Table III. The baseband waveform data samples of the CM signal modulated by the k^{th} CM-scheme

[3]Note that "kB" is the acronym of "kilo-bytes".

TABLE III
THE SPECIFICATIONS OF THE COMPONENTS IN OUR HARDWARE PLATFORM

Instrument	Model	Make	Function
Vector Signal Generator (VSG)	N5182B	Keysight	Generate the CM signals using the CM-schemes in $\mathbb{K}$ at different SNRs.
Signal Analyzer (SA)	N9030B	Keysight	Monitor the CM signals generated by the VSG.
Wideband Digital Receiver (WDR)	M9703B	Keysight	Continuously digitize the CM signal and transmit the acquired CM signal data stream to the workstation for storage.
	M9505A	Keysight	
Workstation	Z8	HP	Seamlessly store the CM signal data stream acquired by the WDR.
Power Splitter	11667A	Keysight	Distribute the CM signal to WDR and SA in parallel.

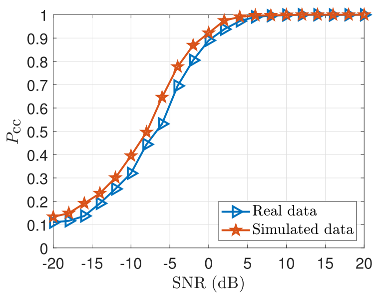

Fig. 18. Probabilities of correct classification, P_{cc}'s, versus SNR for the given CM-scheme candidate set $\mathbb{K}$ resulting from the real CM signal data and the simulated CM signal data in the presence of an AWGN channel.

$\mathscr{K}_k \in \mathbb{K}$ are produced in MATLAB according to the parameters presented in Table I and fed to the VSG to generate the corresponding analog RF CM signal. Here, the center frequency of the RF CM signal is set to be the IF 70 MHz for emulating the IF sampling in a real CM receiver while the AWGN subject to various power levels are injected by the VSG to output the CM signals with respect to different SNRs. Then, the generated analog CM signal is further transmitted to the SA and the WDR in parallel by a power splitter for monitoring and acquisition, respectively. In the WDR, the input analog CM signal is continuously digitized into 12-bit samples at the sampling rate 200 MSps, which are further streamed to the workstation through the PCIe bus for seamlessly storage. Finally, the workstation stores a fixed number of digital CM signal samples as a file at a time for ACMC.

Once all CM signal data modulated by each CM scheme of the candidate set $\mathbb{K}$ subject to each SNR condition are acquired by the hardware platform shown by Fig. 17, they are first digital down-converted and decimated using MATLAB and then fed to our proposed new ACMC scheme for ACMC. The probabilities of correct classification, P_{cc}'s, for the given CM-scheme candidate set $\mathbb{K}$ resulting from the real CM signal data and the simulated CM signal data in the presence of an AWGN channel are delineated by Fig. 18 for comparison. According to Fig. 18, our proposed new ACMC technique can perform very closely over both simulated and real signal data. It can be expected that the performance of our proposed new ACMC technique for real signal data would be slightly worse than that for simulated data sometimes. The SNR margin of our proposed new ACMC technique between simulated and real signal data to perform the same P_{cc} is around 1-2 dB. When SNR$\geq$5 dB, our proposed new ACMC scheme leads to the perfect performance for both real and simulated signal data.

V. CONCLUSION

The automatic composite-modulation (CM) classification (ACMC) is investigated in this work. We propose to first obtain the three-dimensional normalized cyclic spectrum of the received signal. Then the spectrum is transformed into the cyclic paw-print (CPP), which turns out to be a gray-scale image matrix. The CPP is further transformed using the discrete cosine transform and discrete wavelet transform to construct our proposed new hybrid feature vector. According to the hybrid feature vectors obtained from the training and test CM signals, the CM scheme of a received signal can be identified using the multiclass support vector machine. The results from both Monte Carlo simulations and real experiments are also presented to justify the superiority of our proposed new ACMC technique to the existing ACMC methods. Our proposed new robust ACMC technique would be a promising solution to the next generation intelligent telemetry, tracking, and command (TT&C) systems for cognitive space communications and space surveillance.

REFERENCES

[1] C. Zhang, C. Jiang, L. Kuang, J. Jin, Y. He, and Z. Han, "Spatial spectrum sharing for satellite and terrestrial communication networks," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 3, pp. 1075–1089, Jun. 2019.

[2] H. Xie, Y. Zhan, G. Zeng, and X. Pan, "LEO mega-constellations for 6G global coverage: Challenges and opportunities," *IEEE Access*, vol. 9, pp. 164223–164244, 2021.

[3] L. Lu, W. Xu, Y. Wang, and Z. Tian, "Compressive spectrum sensing using sampling-controlled block orthogonal matching pursuit," *IEEE Trans. Commun.*, vol. 71, no. 2, pp. 1096–1111, Feb. 2023.

[4] W. Zhao, S. S. Ali, M. Jin, G. Cui, N. Zhao, and S.-J. Yoo, "Extreme eigenvalues-based detectors for spectrum sensing in cognitive radio networks," *IEEE Trans. Commun.*, vol. 70, no. 1, pp. 538–551, Jan. 2022.

[5] J. Yuan, Y.-C. Liang, J. Joung, G. Feng, and E. G. Larsson, "Intelligent reflecting surface-assisted cognitive radio system," *IEEE Trans. Commun.*, vol. 69, no. 1, pp. 675–687, Jan. 2021.

[6] J. Perez, J. Via, L. Vielva, and D. Ramírez, "Online detection and SNR estimation in cooperative spectrum sensing," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2521–2533, Apr. 2022.

[7] C. Zeng, J.-B. Wang, C. Ding, M. Lin, and J. Wang, "MIMO unmanned surface vessels enabled maritime wireless network coexisting with satellite network: Beamforming and trajectory design," *IEEE Trans. Commun.*, vol. 71, no. 1, pp. 83–100, Jan. 2023.

[8] Y. Hai, Y. Luo, C. Liu, and A. Dang, "Remote phase-shift LiDAR with communication," *IEEE Trans. Commun.*, vol. 71, no. 2, pp. 1059–1070, Feb. 2023.

[9] X. Hao, Z. Feng, S. Yang, M. Wang, and L. Jiao, "Automatic modulation classification via meta-learning," *IEEE Internet Things J.*, vol. 10, no. 14, pp. 12276–12292, Jul. 2023.

[10] W. Anwar, A. Kumar, N. Franchi, and G. Fettweis, "Physical layer performance modeling of modern multicarrier modulation techniques," *IEEE Trans. Commun.*, vol. 70, no. 6, pp. 3725–3741, Jun. 2022.

[11] J. Liu and X. Wang, "Sparsity-exploiting blind receiver algorithms for unsourced multiple access in MIMO and massive MIMO channels," *IEEE Trans. Commun.*, vol. 69, no. 12, pp. 8055–8067, Dec. 2021.

[12] X. Li et al., "Physical-layer authentication for ambient backscatter-aided NOMA symbiotic systems," *IEEE Trans. Commun.*, vol. 71, no. 4, pp. 2288–2303, Apr. 2023.

[13] S. Huang, C. Lin, W. Xu, Y. Gao, Z. Feng, and F. Zhu, "Identification of active attacks in Internet of Things: Joint model- and data-driven automatic modulation classification approach," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 2051–2065, Feb. 2021.

[14] P. Ghasemzadeh, M. Hempel, and H. Sharif, "GS-QRNN: A high-efficiency automatic modulation classifier for cognitive radio IoT," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9467–9477, Jun. 2022.

[15] R. Zhang, S. Chang, Z. Wei, Y. Zhang, S. Huang, and Z. Feng, "Modulation classification of active attacks in Internet of Things: Lightweight MCBLDN with spatial transformer network," *IEEE Internet Things J.*, vol. 9, no. 19, pp. 19132–19146, Oct. 2022.

[16] D. Zhu, V. J. Mathews, and D. H. Detienne, "A likelihood-based algorithm for blind identification of QAM and PSK signals," *IEEE Trans. Wireless Commun.*, vol. 17, no. 5, pp. 3417–3430, May 2018.

[17] Y. A. Eldemerdash, O. A. Dobre, O. Üreten, and T. Yensen, "A robust modulation classification method for PSK signals using random graphs," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 2, pp. 642–644, Feb. 2019.

[18] Y. Yuan, P. Zhao, B. Wang, and B. Wu, "Hybrid maximum likelihood modulation classification for continuous phase modulations," *IEEE Commun. Lett.*, vol. 20, no. 3, pp. 450–453, Mar. 2016.

[19] A. O. Abdul Salam, R. E. Sheriff, Y.-F. Hu, S. R. Al-Araji, and K. Mezher, "Automatic modulation classification using interacting multiple model Kalman filter for channel estimation," *IEEE Trans. Veh. Technol.*, vol. 68, no. 9, pp. 8928–8939, Sep. 2019.

[20] C.-F. Teng, C.-Y. Chou, C.-H. Chen, and A.-Y. Wu, "Accumulated polar feature-based deep learning for efficient and lightweight automatic modulation classification with channel compensation mechanism," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 15472–15485, Dec. 2020.

[21] S. Lin, Y. Zeng, and Y. Gong, "Modulation recognition using signal enhancement and multistage attention mechanism," *IEEE Trans. Wireless Commun.*, vol. 21, no. 11, pp. 9921–9935, Nov. 2022.

[22] J. Li et al., "Composite multiple-mode orthogonal frequency division multiplexing with index modulation," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 3748–3761, Jun. 2023.

[23] D. Shuli, L. Zhipeng, and Z. Linfeng, "A modulation recognition algorithm based on cyclic spectrum and SVM classification," in *Proc. IEEE 4th Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, vol. 1, Jun. 2020, pp. 2123–2127.

[24] M. H. Moghaddam, S. R. Aghdam, N. Mazzali, and T. Eriksson, "Statistical modeling and analysis of power amplifier nonlinearities in communication systems," *IEEE Trans. Commun.*, vol. 70, no. 2, pp. 822–835, Feb. 2022.

[25] L. Wang, Y. Han, P. Zhou, H. Ge, and L. Shen, "Modulation recognition of composite modulation signal based on two-fold digital receiver and goodness of fit test," in *Proc. IEEE Int. Conf. Electron. Technol., Commun. Inf. (ICETCI)*, Aug. 2021, pp. 433–437.

[26] F. Li, Z. Yang, B. Huang, and Y. Chen, "Composite radar modulation identification by transfer learning," in *Proc. Int. Conf. Microw. Millim. Wave Technol. (ICMMT)*, May 2021, pp. 1–3.

[27] P. Zhao, J. Gao, and T. Wang, "Parameter estimation of LFM-BPSK composite modulated radar signal based on multi-output regression network," in *Proc. Int. Symp. Antennas Propag. (ISAP)*, Oct. 2022, pp. 379–380.

[28] C. Chang, Z. Li, G. Yao, and Y. Xia, "Adaptive composite phases modulation/demodulation based on transform domain communication system," in *Proc. IEEE 4th Int. Conf. Comput. Commun. (ICCC)*, Dec. 2018, pp. 52–56.

[29] P. Neumaier, M. Bochenko, A. Blumenschein, N. Hansen, J. Lex, and S. Lakshmaiah, "Multiplex modulation techniques for a navigation constellation simulator," in *Proc. IEEE/ION Position, Location Navigat. Symp. (PLANS)*, Apr. 2018, pp. 1428–1438.

[30] A. K. Karthik and R. S. Blum, "Robust clock skew and offset estimation for IEEE 1588 in the presence of unexpected deterministic path delay asymmetries," *IEEE Trans. Commun.*, vol. 68, no. 8, pp. 5102–5119, Aug. 2020.

[31] P. Chu, L. Xie, C. Dai, and Y. Chen, "Automatic modulation recognition for secondary modulated signals," *IEEE Wireless Commun. Lett.*, vol. 10, no. 5, pp. 962–965, May 2021.

[32] Q. Xu, J. Wan, L. He, and Q. Wang, "Blind recognition for hybrid modulation signals based on high-order cumulant and squared spectrum feature," *Acta Electronica Sinica*, vol. 50, no. 2, pp. 426–431, Feb. 2022.

[33] T. Ravi Kishore and K. D. Rao, "Automatic intrapulse modulation classification of advanced LPI radar waveforms," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 2, pp. 901–914, Apr. 2017.

[34] X. Wu, S. Wei, Y. Zhou, and F. Liao, "TSN-A: An efficient deep learning model for automatic modulation classification based on intra-class confusion reduction of modulation families," *IEEE Commun. Lett.*, vol. 26, no. 12, pp. 2964–2968, Dec. 2022.

[35] X. Mao, J. Gao, and J. Qi, "Modulation recognition algorithm of radar signal based on ICanny-CNN," in *Proc. Int. Symp. Antennas Propag. (ISAP)*, Oct. 2021, pp. 1–2.

[36] L. Li, Z. Dong, Z. Zhu, and Q. Jiang, "Deep-learning hopping capture model for automatic modulation classification of wireless communication signals," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 772–783, Apr. 2023.

[37] S. Huang et al., "Automatic modulation classification using gated recurrent residual network," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7795–7807, Aug. 2020.

[38] S. Kojima, K. Maruta, Y. Feng, C.-J. Ahn, and V. Tarokh, "CNN-based joint SNR and Doppler shift classification using spectrogram images for adaptive modulation and coding," *IEEE Trans. Commun.*, vol. 69, no. 8, pp. 5152–5167, Aug. 2021.

[39] T. Huynh-The, T. Nguyen, Q. Pham, D. B. Da Costa, G. Kwon, and D. Kim, "Efficient convolutional networks for robust automatic modulation classification in OFDM-based wireless systems," *IEEE Syst. J.*, vol. 17, no. 1, pp. 964–975, Mar. 2023.

[40] J. Ma, S.-C. Lin, H. Gao, and T. Qiu, "Automatic modulation classification under non-Gaussian noise: A deep residual learning approach," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2019, pp. 1–6.

[41] I. Lee and W. Lee, "UniQGAN: Unified generative adversarial networks for augmented modulation classification," *IEEE Commun. Lett.*, vol. 26, no. 2, pp. 355–358, Feb. 2022.

[42] Z. Wang, W. Liu, and H.-M. Wang, "GAN against adversarial attacks in radio signal classification," *IEEE Commun. Lett.*, vol. 26, no. 12, pp. 2851–2854, Dec. 2022.

[43] A. Modenini and B. Ripani, "A tutorial on the tracking, telemetry, and command (TT&C) for space missions," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 3, pp. 1510–1542, 3rd Quart., 2023.

[44] J. Zhao, W. Yang, and Y. Cheng, "Exploration of cognitive space TT&C system," in *Proc. 12th Int. Conf. Signal Process. (ICSP)*, Oct. 2014, pp. 1739–1742.

[45] *Radio Frequency and Modulation Systems—Part 1 Earth Stations and Spacecraft*, Consultive Committee for Space Data Syst. (CCSDS), Washington, DC, USA, Feb. 2020.

[46] Y. Zhan, P. Wan, C. Jiang, X. Pan, X. Chen, and S. Guo, "Challenges and solutions for the satellite tracking, telemetry, and command system," *IEEE Wireless Commun.*, vol. 27, no. 6, pp. 12–18, Dec. 2020.

[47] X. Yan, G. Feng, H. Wu, W. Xiang, and Q. Wang, "Innovative robust modulation classification using graph-based cyclic-spectrum analysis," *IEEE Commun. Lett.*, vol. 21, no. 1, pp. 16–19, Jan. 2017.

[48] X. Yan, G. Liu, H. Wu, and G. Feng, "New automatic modulation classifier using cyclic-spectrum graphs with optimal training features," *IEEE Commun. Lett.*, vol. 22, no. 6, pp. 1204–1207, Jun. 2018.

[49] X. Yan, G. Liu, H.-C. Wu, G. Zhang, Q. Wang, and Y. Wu, "Robust modulation classification over α-stable noise using graph-based fractional lower-order cyclic spectrum analysis," *IEEE Trans. Veh. Technol.*, vol. 69, no. 3, pp. 2836–2849, Mar. 2020.

[50] F. Cai and V. Cherkassky, "Generalized SMO algorithm for SVM-based multitask learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 6, pp. 997–1003, Jun. 2012.

[51] H.-K. Le, V.-P. Hoang, V.-S. Doan, M.-T. Hoang, and N. P. Dao, "Performance analysis of convolutional neural networks with different window functions for automatic modulation classification," in *Proc. 13th Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, Oct. 2022, pp. 153–157.

[52] D. D. Wackerly, W. Mendenhall III, and R. L. Scheaffer, *Mathematical Statistics With Applications*, 7th ed. Pacific Grove, CA, USA: Brooks/Cole, 2008.

[53] D. S. Soper, "Greed is good: Rapid hyperparameter optimization and model selection using greedy k-Fold cross validation," *Electronics*, vol. 10, no. 16, p. 1973, Aug. 2021.

[54] F. Gardner, "A BPSK/QPSK timing-error detector for sampled receivers," *IEEE Trans. Commun.*, vol. COM-34, no. 5, pp. 423–429, May 1986.

[55] P. W. Kinman and J. B. Berner, "Carrier synchronization of offset QPSK for deep space telemetry," in *Proc. IEEE Aerosp. Conf.*, Mar. 2002, p. 3.

[56] M. Bertolucci, R. Cassettari, and L. Fanucci, "CCSDS 131.2-B-1 frequency estimation trade-offs and a novel multi-algorithm FPGA architecture," in *Proc. Int. Conf. Comput., Electron. Commun. Eng. (iCCECE)*, Aug. 2020, pp. 224–229.

[57] G. Meoni et al., "CCSDS 131.2-B-1 telemetry transmitter: A VHDL IP core and a validation architecture on board RTG4 FPGA," *Acta Astronautica*, vol. 176, pp. 484–493, Nov. 2020.

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